



T.C.
DOKUZ EYLÜL UNIVERSITY
ENGINEERING FACULTY
ELECTRICAL & ELECTRONICS ENGINEERING
DEPARTMENT



Design Of A Three-Phase Inverter Drive With Space Vector PWM Technique

Final Project

by

Mehmet Selim Keleş
2016502062

Advisor

Assist. Prof. Dr. Tolga SÜRGEVİL

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THESIS EVALUATION FORM

We certify that we have read this thesis and that in our opinion it is fully adequate, in scope and qualify as an undergraduate thesis, based on the result of the oral examination taken place on 02/06/2022

Assist. Prof. Dr. Tolga SÜRGEVİL
(ADVISOR)

Prof. Dr. Eyüp AKPINAR Dr. Taner GÖKTAŞ
(COMMITTEE MEMBER) (COMMITTEE MEMBER)

Prof Dr. Mehmet KUNTALP
(CHAIRPERSON)

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ABSTRACT

This project aims to design and simulation of the 3-Phase voltage source inverter with the Space Vector Pulse Width Modulation(SVPWM) technique. For this purpose both Conventional and Carrier Based Space Vector Modulation techniques have been examined and 2 different designs created. 2 Level Pulse Width Modulated-Voltage Source Inverter(PWM-VSI) topology and a three-phase RL load used for the hardware implementation of the project. In Conventional SVPWM reference signals for the PWM-VSI circuit have been obtained by using Clarke Transformation. For Carrier Based SVPWM model, modified three-phase sinusoids have been created for reference signals using real-time relations. Results and analysis of the simulation model made by using MATLAB-Simulink software compared to the calculated values.

Implementation of the model made by using STM32F407VGT6 microcontroller as Digital Signal Processor. Also, the VSI circuit designed in KiCAD software and printed for hardware implementation. The design philosophy of the Printed Circuit Board and the algorithmic structure of the DSP are explained in detail. In the implementation, the Conventional SVPWM method is preferred. 3-Phase induction motor used as a load for measurements and observations. Cost analysis of the system has shared respectively.

ÖZET

Bu proje, Uzay Vektör Darbe Genişlik Modülasyonu (SVPWM) tekniği ile 3-Fazlı gerilim kaynaklı evirici tasarımı ve simülasyonunu amaçlamaktadır. Bu amaçla hem Konvansiyonel hem de Taşıyıcı Tabanlı Uzay Vektör Modülasyon teknikleri incelenmiş ve 2 farklı tasarım oluşturulmuştur. Projenin donanım kısmı için Seviye 2 Darbe Genişliği Modülasyonlu-Gerilim Kaynak Çevirici (PWM-VSI) topolojisi ve üç fazlı RL yükü kullanılmıştır. Konvansiyonel SVPWM'de PWM-VSI devresi için referans sinyalleri Clarke Dönüşümü kullanılarak elde edilmiş ve bu sinyalleri kullanarak da anahtarlama sinyalleri elde edilmiştir. Taşıyıcı Tabanlı SVPWM modeli için, 3-faz sinyaller arasındaki gerçek zamanlı ilişkiler kullanılarak modifiye edilmiş üç fazlı referans sinyalleri oluşturulmuştur. MATLAB-Simulink yazılımı kullanılarak yapılan simülasyon modellenin; sonuçları ve analizi, hesaplanan değerlerle karşılaştırılmıştır.

Modelin gerçekleştirilmesinde Sayısal Sinyal İşlemcisi olarak STM32F407VGT6 mikroişlemcisi kullanılmıştır. Ayrıca donanım kısmını gerçeklemek için bir VSI devresi KiCAD yazılımı kullanarak tasarlanmış ve basılmıştır. Baskıdevrenin ve DSP algoritmasının tasarım felsefesi detaylıca anlatılmıştır. Gerçekleme için Konvansiyonel SVPWM metodu tercih edilmiştir. Ölçümler ve gözlemler için 3-Fazlı endüksiyon motoru kullanılmıştır. Sistem maliyet analizi sırayla paylaşılmıştır.

Contents

ACKNOWLEDGEMENTS	I
ABSTRACT	II
ÖZET	III
Contents	IV
List of Tables	VI
List of Figures	VII
1 INTRODUCTION	1
2 Literature Review on 3 Phase Voltage Source Inverters	3
2.1 3 Phase Voltage Source Inverter Circuit	3
2.2 Switching techniques in 3-phase VSI circuits	4
2.2.1 180° Conduction	4
2.2.2 120° Conduction	7
2.2.3 Sinusoidal Pulse Width Modulation	8
2.2.4 3rd Harmonic Pulse Width Modulation	10
2.3 Space Vector Pulse Width Modulation	11
2.3.1 Clarke Transformation	11
2.3.2 State Vectors and Sectors	12
2.3.3 Obtaining Gate Signals	13
2.3.4 Modulation Index and Linear Region	18
3 Simulation of the System	19
3.1 Simulation Model	19
3.1.1 Conventional Modulator	20
3.1.2 Carrier Based Modulator	23
3.1.3 Inverter Circuit	24
3.2 Simulation Results	25

3.2.1	Gate Reference Signals and Gate Signals	25
3.2.2	Phase Currents and Phase Voltages	26
3.2.3	FFT Results	28
4	Implementation of the System	30
4.1	Hardware Implementation	30
4.1.1	Schematic Design of the Circuit	30
4.1.2	Printed Circuit Board Design of the Circuit	34
4.2	Software Implementation	36
5	Experimental Results and Discussions	38
5.1	Hardware Results	38
5.2	Software Results	39
5.3	Experiment Setup	39
5.4	Experiment Results and Comparision with Theoric Calculations	41
5.4.1	Results at 39Hz	41
5.4.2	Operation at Various Frequencies	43
5.5	FFT Results of the Measurements	44
6	Cost Analysis	45
6.1	Economical Costs	45
6.2	Environmental, Political and Social Costs	45
7	Conclusion	46
8	References	47

List of Tables

2.1	Parameters for Inverter State	5
2.2	State Vectors With Clarke Transform	13
2.3	State Switching Sequences	15
2.4	Conduction Times for High Side Switches Over 1 Period	16
6.1	Cost Table	45



List of Figures

2.1	Voltage Source Inverter Circuit	3
2.2	Gate Signals for 180° conduction	4
2.3	Equivalent circuits for each state with balanced 3 phase load	5
2.4	Waveforms for 180° conduction	6
2.5	Gate Signals for 120° conduction	7
2.6	Waveforms 120° conduction	8
2.7	Comparison of Reference Signal to Carrier, Gate Signals, Phase Voltages and Line Voltages	9
2.8	Visualization of 3rd Harmonic Injection to the Reference Signal	10
2.9	Visualization of Clarke Transform	11
2.10	Visualization of the State Vectors and Sectors with Clarke Transform	13
2.11	Illustrated Vector Addition in SVPWM	14
2.12	Highly Sampled Conduction Time Data Plot	16
2.13	Maximum Minimum Detection with Real Time Sampling for 3-Phase System	17
2.14	Geometrical Derivation of $\vec{V}_{r,max}$	18
3.1	Simulink Model	19
3.2	Conventional Modulator Structure	20
3.3	Sector Calculator Structure	21
3.4	Dwell Time Calculator Structure	22
3.5	Sequence Generator Structure	22
3.6	Carrier Based Modulator Structure	23
3.7	Max-Min Generator Block	24
3.8	3-Phase PWM-VSI Circuit	24
3.9	Gate Reference Signals	25
3.10	Gate Signals in Sector 1	26
3.11	Phase Voltages	26
3.12	Phase Currents	27
3.13	V_{AN} , V_{Nn} and V_{An}	27

3.14	FFT Result for Phase Voltage	28
3.15	FFT Result for V_{AN}	29
3.16	FFT Result for V_{Nn}	29
4.1	Schematic Drawing of the Circuit	31
4.2	Gate Drive Circuit	32
4.3	PWM-VSI Circuit	33
4.4	3d Preview of the PCB	34
4.5	PCB Drawing of the Circuit	35
4.6	Flowchart of the Software	37
5.1	Rise and Fall time for IGBTs' Gate	38
5.2	Switching signals for one gate after applying low-pass filter	39
5.3	Nameplate of the AC motor used in experiments	40
5.4	Experimental Setup	40
5.5	Phase Voltages at 39Hz	41
5.6	Phase Currents at 39Hz	42
5.7	Line-to-Line Voltages at 39Hz	42
5.8	Operation at 21Hz	43
5.9	Operation at 59Hz	43
5.10	FFT Results in shape of Bar Graph	44

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1. INTRODUCTION

Demanded variable frequency and variable amplitude for AC machines also discoveries and improvements on chemical energy storage units which stores DC power, opened up new discovery areas in electronics for DC-AC conversion. We use the engineering term Inverter” for these special circuits. Today, the IEEE dictionary similarly defines Inverter (electric power) as ”a machine, device, or system that changes direct-current power to alternating-current power”. Inverters divide into two sections called Voltage Source Inverters(VSI) and Current Source Inverters(CSI). Pulse Width Modulated Voltage Source Inverters widely used in industry since the development of thyristors and controllable semiconductors in the 1960s. There are many topologies for different circumstances that have advantages and disadvantages to each other. Also, Inverters can be fed from rectified AC sources since the grid signal has a fixed frequency and amplitude that is decided by distributors.

Pulse Width Modulated Voltage Source Inverters(PWM-VSI) become much more efficient and practical to use since the invention of IGBTs. IGBTs are a special kind of semiconductor that consists of advantages over both MOSFETs and BJTs. IGBTs have very low internal resistance like BJTs also they don’t need huge gate currents like MOSFETs. Besides this; for the same VSI topologies, different modulation techniques have been invented e.g. Sinusoidal PWM, Third Harmonic PWM, and Space Vector PWM. The difference between these modulation techniques is derived from switching pattern differences which result in differences in DC bar utilization and design complexity. They have both advantages and disadvantages. For example, Sinusoidal PWM has an easier design procedure but less DC Bus utilization compared to others, also Space Vector Modulation has a more complex design but this complexity results in less harmonic distortion compared to the Third Harmonic PWM [1].

Space Vector Modulation contains a huge place over these methods which is also the topic of this thesis. In 1982 conference paper by Pfaff, Weschta and Wick described creating switching waveforms utilizing a technique for electric machine analysis known as space vectors [2]. The basic idea is that you analyze the state of a motor or generator by considering per-phase quantities together as a group. This idea had been around since the early 1900s but was primarily developed by R.H. Park at General Electric, who published a 1929 paper on the so-called "two-reaction" (d-q) theory [3]. The term "space vector" appears to be of European origin (by two Hungarian professors, K. P. Kovacs and I. Racz), referring to the use of symmetrical components and Park's two-reaction theory, not just as abstract quantities used to analyze three-phase power, but as real physical quantities representing the spatial variation of currents or magnetic fluxes in electric motors or generators. The technique used by Pfaff, Weschta, and Wick for generating switching waveforms is to treat 3-Phase PWM generation as a single task. Instead of generating three independent switching waveforms, treat the 3-Phase bridge as one unit. A 1986 conference paper by Van Der Broeck, Skudelny, and Stanke described the "optimal" use of space vector modulation in PWM to reduce switching losses and current ripple. The Van Der Broeck paper has been cited by over 600 papers about motor control and is really the foundation for PWM in modern digital motor control [4].

It must be noted that Space Vector Modulation is much more a general idea and does not need any fixed frequency PWM. But technique mentioned in Van Der Broeck's papers uses fixed frequency and is called Conventional SVPWM. [5] When someone refers to Space Vector Modulation usually means Van Der Broeck's method.

This thesis contains an Inverter design with Space Vector PWM. In the Literature Review chapter, 3-Phase PWM-VSI topology was examined. Also popular and widely used switching methods and techniques with Space Vector PWM are reviewed in a section under the chapter. Simulation Model and Results are shared in the Simulation of the System chapter. Implementation of the model and a detailed explanation of the design are shared in the Implementation of the System chapter. Measurements and observations of the implemented model, also comparisons of simulation results can be found in the Experimental Results chapter. Cost Analysis and Conclusion of the thesis may found in order.

2. Literature Review on 3 Phase Voltage Source Inverters

In this chapter, analysis is made on 2 Level 3-Phase PWM-VSI circuit since it is the same topology that used in this thesis model and also shared by other modulation techniques too. Also a couple of switching techniques that help to better understand Space Vector PWM examined. 180° and 120° Conduction methods are widely used in the industry throughout history for their ease of usability. Also, very common techniques Sinusoidal PWM and slightly different 3rd Harmonic PWM mentioned respectively. 3rd Harmonic PWM's philosophy and its waveforms plays a huge role in understanding Space Vector PWM since the development of both techniques is aimed at the same goal.

2.1 3 Phase Voltage Source Inverter Circuit

Three Phase VSI(Voltage Source Inverter) topology is given in figure 2.1 below. 6 Switch with 3 legs can be seen at this topology. We classify the leg's upper switches as the high side and lower switches as the low side. A leg's high side and low side switches can not be closed at the same time due to the possible shorting of DC Source. Freewheeling Diodes are used with every IGBT to prevent switch time delays or damage on IGBTs while turning off.

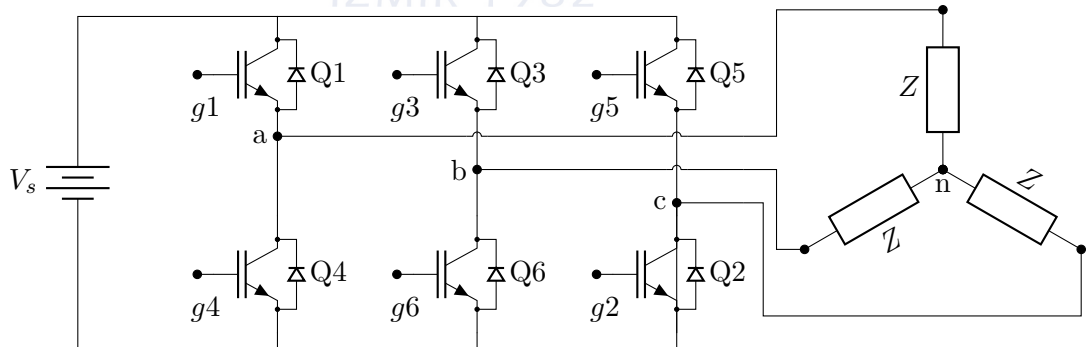


Figure 2.1: Voltage Source Inverter Circuit

DC Source's Positive Terminal is a common connection for all legs' high sides and the negative terminal is common ground for low sides. Outputs are considered the midpoint of every leg. Since there is 3 leg, 3 signal can be generated. Since each terminal can only be connected to the two potentials (with respect to the bus midpoint) this is called a two-level inverter. With the help of control signals for switches, parameters for generated signals (Amplitude and Frequency) can be adjusted.

2.2 Switching techniques in 3-phase VSI circuits

There are many switching techniques out there when the topic is 3-phase PWM-VSI circuits. But in this chapter, widely used ones which play a role in the development of modulation techniques are examined. Respectively 180° and 120° Conduction techniques are examined first since they exist from the start of PWM-VSI topology, also they give a fundamental idea of PWM-VSI. After that, more complex techniques like Sinusoidal PWM and 3rd Harmonic Injection are reviewed for clarification of Space Vector PWM.

2.2.1 180° Conduction

In 180° conduction gate signals applied as high logic state in the first half period which closes the high side switch. And in the second half of the period low logic state is applied which opens the high side switch. The low side switch is the complement of the high side switch. Since High Side conducting for 180° technique is called 180° Conduction. To obtain 3-Phase and 120° phase difference between output signals, same control signals with phase difference -120° and 120° applied to the B and C legs. Visualization of gate signals for all switches can be seen at figure 2.2

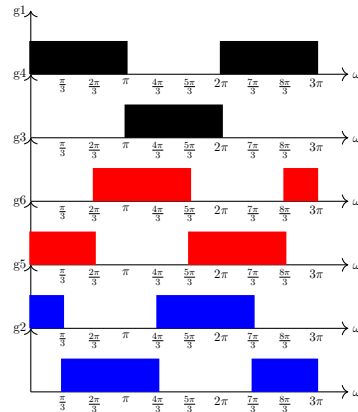


Figure 2.2: Gate Signals for 180° conduction

This technique creates 6 different circuits in the switching sequence. To analyze them easily we can use notation where the legs' high side closed as 1 and the low side closed as 0. All 6 circuit's for balanced 3-Phase load shown in figure 2.3. Also for switching states instant voltage parameters are filled in table 2.1.

Table 2.1: Parameters for Inverter State

State	Conducting Switches	$V_{\phi a}$	$V_{\phi b}$	$V_{\phi c}$	V_{ab}	V_{bc}	V_{ca}
101	Q1, Q6, Q5	$\frac{1}{3}V_s$	$-\frac{2}{3}V_s$	$\frac{1}{3}V_s$	V_s	$-V_s$	0
100	Q1, Q6, Q2	$\frac{2}{3}V_s$	$-\frac{1}{3}V_s$	$-\frac{1}{3}V_s$	V_s	0	$-V_s$
110	Q1, Q3, Q2	$\frac{1}{3}V_s$	$\frac{1}{3}V_s$	$-\frac{2}{3}V_s$	0	V_s	$-V_s$
010	Q4, Q3, Q2	$-\frac{1}{3}V_s$	$\frac{2}{3}V_s$	$-\frac{1}{3}V_s$	$-V_s$	V_s	0
011	Q4, Q3, Q5	$-\frac{2}{3}V_s$	$\frac{1}{3}V_s$	$\frac{1}{3}V_s$	$-V_s$	0	V_s
001	Q4, Q6, Q5	$-\frac{1}{3}V_s$	$-\frac{1}{3}V_s$	$\frac{2}{3}V_s$	0	V_s	V_s
111	Q1, Q3, Q5	0	0	0	0	0	0
000	Q4, Q6, Q2	0	0	0	0	0	0

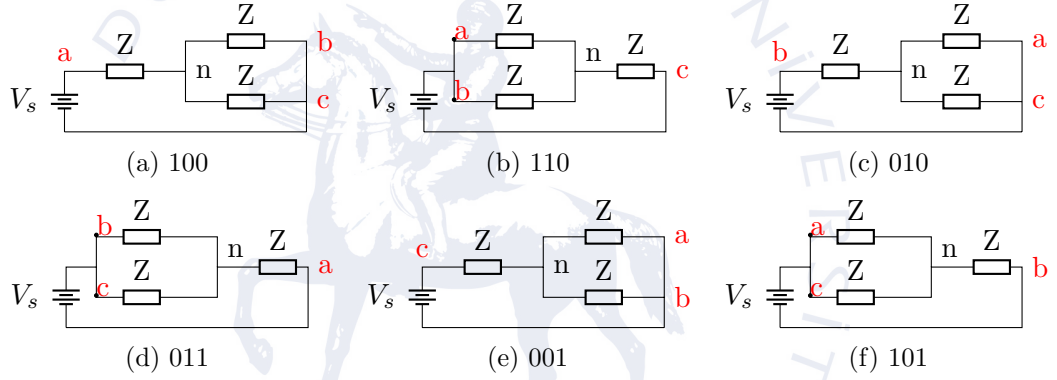


Figure 2.3: Equivalent circuits for each state with balanced 3 phase load

Obtained Phase and Line to Line Voltages waveforms can be seen at figure 2.4. Mathematical derivation of Line to Line voltages can be found by shifting signal 60° to right and obtaining odd quarter wave symmetry. By Fourier Series Expansion we can write:

$$v_{ab} = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos(n\omega t) + b_n \sin(n\omega t)) \quad (2.1)$$

where a_n and a_0 are zero due to symmetry.

$$b_n = \frac{1}{\pi} \left[\int_{-\frac{5\pi}{6}}^{-\frac{\pi}{6}} -V_s \sin(n\omega t) d(\omega t) + \int_{\frac{\pi}{6}}^{\frac{5\pi}{6}} V_s \sin(n\omega t) d(\omega t) \right] \quad (2.2)$$

$$\Rightarrow v_{ab} = \sum_{n=1,3,5\ldots}^{\infty} \frac{4V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{\pi}{6}\right)\right] \quad (2.3)$$

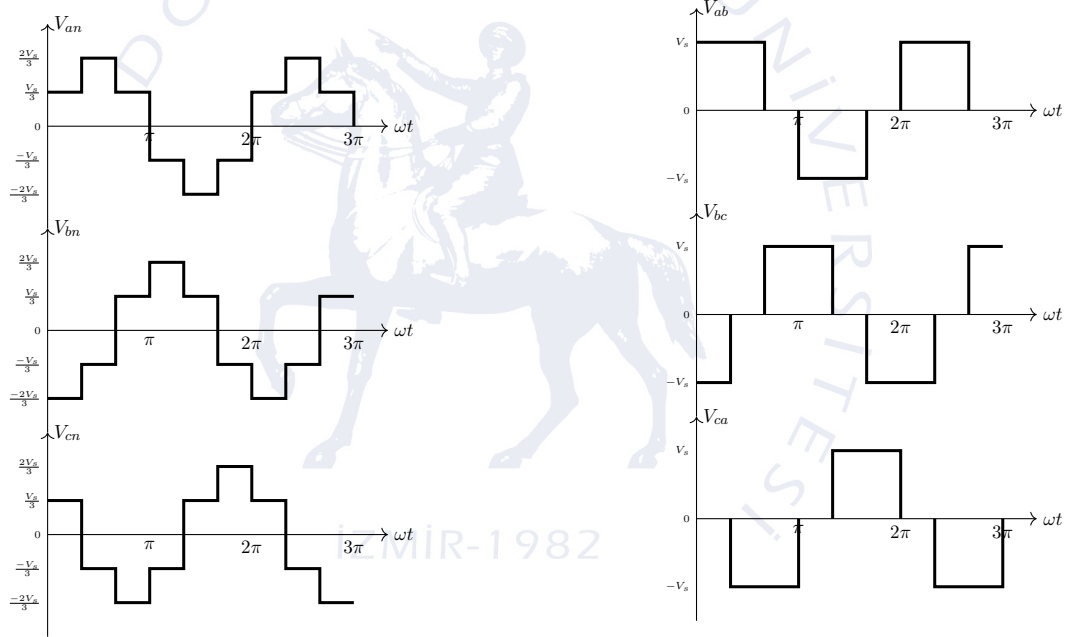
$$\Rightarrow v_{bc} = \sum_{n=1,3,5\ldots}^{\infty} \frac{4V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{-\pi}{2}\right)\right] \quad (2.4)$$

$$\Rightarrow v_{ca} = \sum_{n=1,3,5\ldots}^{\infty} \frac{4V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{-7\pi}{6}\right)\right] \quad (2.5)$$

then we can write Line to Line and Phase RMS voltages as given below.

$$V_{LLrms} = \left[\frac{1}{2\pi} \int_0^{2\pi} V_s^2 d(\omega t) \right]^{\frac{1}{2}} = \sqrt{\frac{2}{3}} V_s = 0.8165 V_s \quad (2.6)$$

$$V_{\phi rms} = \frac{V_{LLrms}}{\sqrt{3}} = 0.4714 V_s \quad (2.7)$$



(a) Phase Voltages for 180° conduction

(b) Line to Line Voltages for 180° conduction

Figure 2.4: Waveforms for 180° conduction

2.2.2 120° Conduction

In 120° conduction high sides and low sides conducts for $\frac{1}{3}T$. There is an off state for each leg which corresponds to $\frac{1}{3}T$ of the period. Since high side conducts for 120° technique is called 120° Conduction. Visualization of gate signals for all legs can be seen in figure 2.5

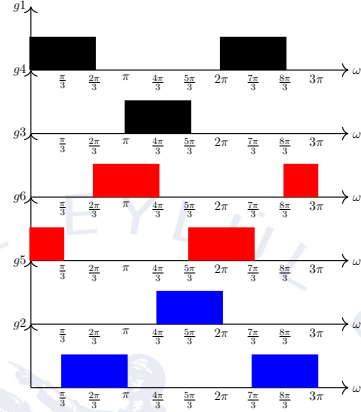


Figure 2.5: Gate Signals for 120° conduction

Obtained Phase and Line to Line Voltages waveforms can be seen in figure 2.6. Mathematical derivation of the Line to Line voltages can be found using Fourier Series Expansion.

$$v_{ab} = \sum_{n=1,3,5\dots}^{\infty} \frac{2\sqrt{3}V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{\pi}{6}\right) \pm \frac{\pi}{6}\right] \quad (2.8)$$

$$v_{bc} = \sum_{n=1,3,5\dots}^{\infty} \frac{2\sqrt{3}V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{-\pi}{2}\right) \pm \frac{\pi}{6}\right] \quad (2.9)$$

$$v_{ca} = \sum_{n=1,3,5\dots}^{\infty} \frac{2\sqrt{3}V_s}{n\pi} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{3}\right) \sin\left[n\left(\omega t + \frac{-7\pi}{6}\right) \pm \frac{\pi}{6}\right] \quad (2.10)$$

$$V_{LLrms} = \frac{1}{\sqrt{2}} = 0.707V_s \quad (2.11)$$

$$V_{\phi rms} = \frac{V_{LLrms}}{\sqrt{3}} = 0.408V_s \quad (2.12)$$

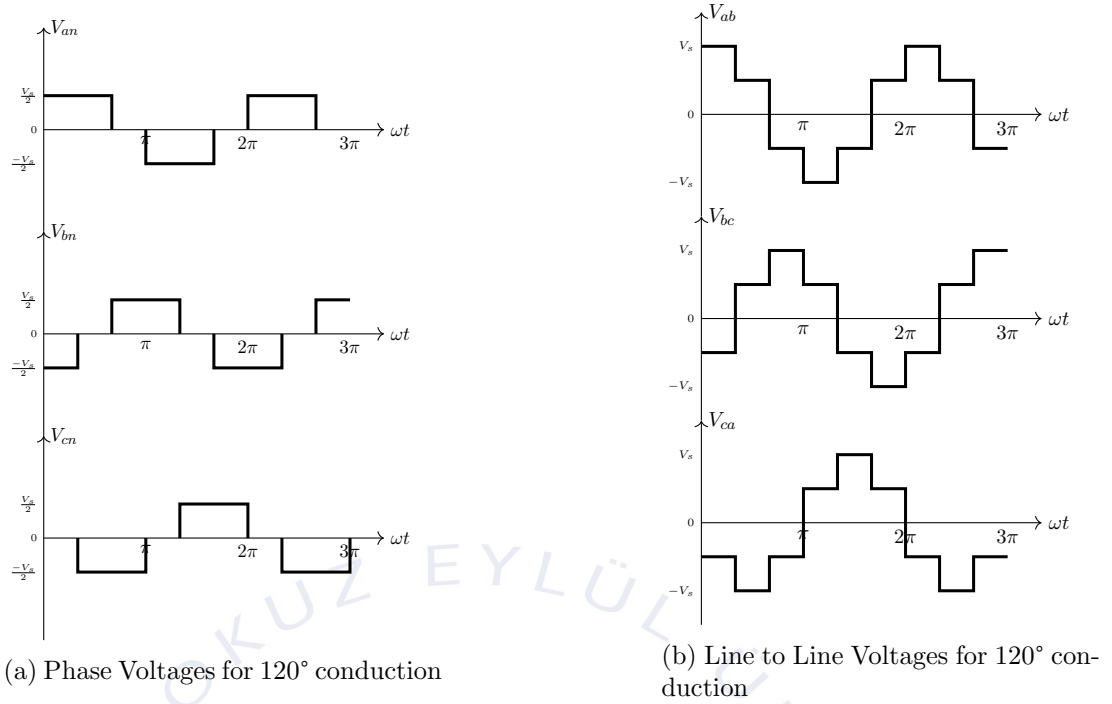
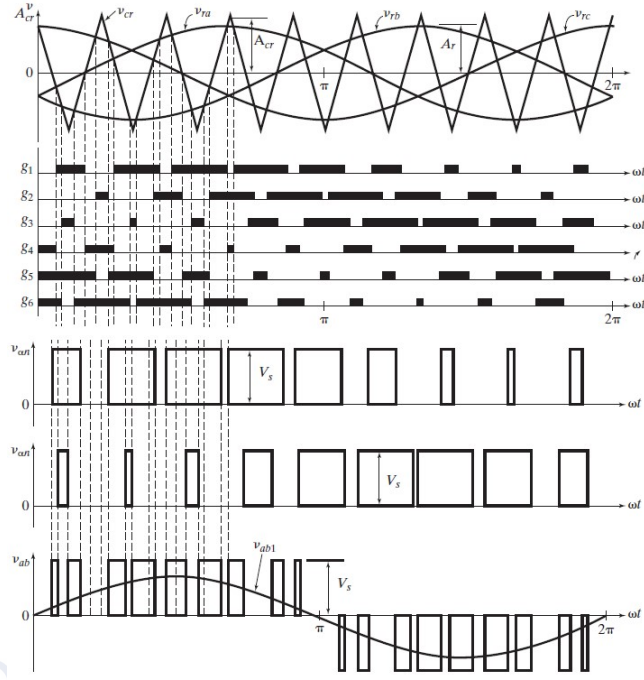


Figure 2.6: Waveforms 120° conduction

2.2.3 Sinusoidal Pulse Width Modulation

Sinusoidal PWM is a technique that uses special gate signals. Pulses for gate signals derived from sine functions. To obtain this waveform, sine function (with desired frequency and normalized amplitudes with respect to DC voltage) compared to triangular wave (carrier) with higher frequency.

There is 3 reference sine function which has 120° phase difference between each other in this modulation technique. Reference signals and carrier signal compared to each other; while the reference signal's amplitude is greater than the carrier signal's amplitude, high side switches are on. Low side switches use a complementary manner like always and they are off. Comparison and generated gate signal for 3-Phase can be seen in figure 2.7 below.



[6]

Figure 2.7: Comparison of Reference Signal to Carrier, Gate Signals, Phase Voltages and Line Voltages

Inverter's output frequency equals to reference signal's frequency. Also, Inverter's output voltage can be adjusted by changing the reference signals' amplitude with respect to the carrier signal's amplitude.

In this modulation technique, there is a parameter called Modulation Index (M) which corresponds to the ratio between reference signal amplitude and carrier amplitude. Linear region specified as $\frac{V_{ref}}{V_{carrier}} = M \leq 1$. In the linear region, the maximum amplitude of the phase to neutral voltage is $\frac{V_s}{2}$. Clearly, the output voltage is equal to the fundamental signal and has very high order harmonics closer to the carrier frequency. Then we can write:

$$V_\phi = M \frac{V_s}{2} \sin(\omega t) \quad (2.13)$$

2.2.4 3rd Harmonic Pulse Width Modulation

3rd Harmonic Pulse Width Modulation (3rd Harmonic PWM) or 3rd Harmonic Injection is a technique similar to the Sinusoidal Pulse Width Modulation. The technique takes its name from the 3rd Harmonic injection to the Reference Signals. By injecting 3rd Harmonic, pit at Sinusoidal Reference's Maximum and Minimum point occurs. With the help of this pitted Reference Signal might be more expandable. As a result reference signal does not exceed DC supply but also fundamental component is higher than the DC supply. Visualization can be seen at figure 2.8 [6]

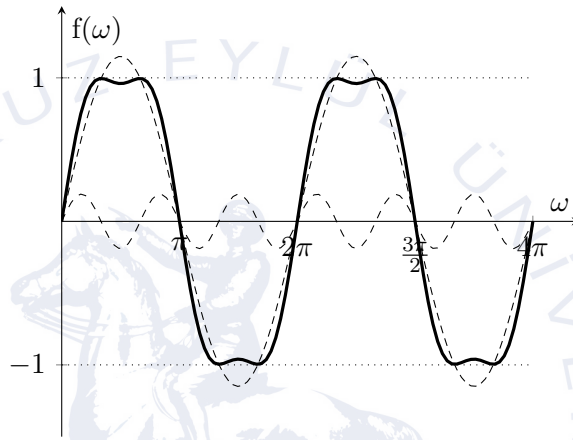


Figure 2.8: Visualization of 3rd Harmonic Injection to the Reference Signal

Since 3rd harmonic was injected on all reference signals it results in the cancellation of the third harmonic component in load's neutral and therefore in phase voltages. When normalized fundamental component expand to $\frac{2}{\sqrt{3}}$, 3rd harmonic component is equals to $\frac{1}{\sqrt{3}}$ of normalized DC supply level. Since fundamental component increased %15 phase voltages increased %15 too which results in higher amplitude for phase voltages with the same DC supply. Can be said 3rd Harmonic Pulse Width Modulation brings higher utilization of DC supply than Sinusoidal Pulse Width Modulation.

$$V_{\phi} = M \frac{2V_s}{\sqrt{3}} \sin(\omega t) \quad (2.14)$$

2.3 Space Vector Pulse Width Modulation

SVPWM (Space Vector Pulse Width Modulation) is a kind of technique where the Inverter's switches are treated as a single unit. Gate Signals are created by using algebraic equations. In SVPWM we consider Reference Voltages as vectors in the 2D plane. Using an algebraic operation called Clarke Transform we obtain 2 Reference Signals with a 90° phase difference. These two signals are AC parameters. Then in one cycle, summation of these to vector creates full circular movement. Gate Signals are created by using this vector's properties.

2.3.1 Clarke Transformation

Clarke transform helps us to represent 3 AC quantities with 2 AC quantities. To use this transformation, a balanced system is needed which is suitable for a 3-Phase System with 120° phase shift between signals. Using α and β parameters we illustrate the 3-Phase system via 2-Phase system with the help of Clarke Transformation. Resultant α and β parameters have 90° phase shift between each other due to selected geometry. Visualization of Clarke Transformation can be seen in figure 2.9 given below.

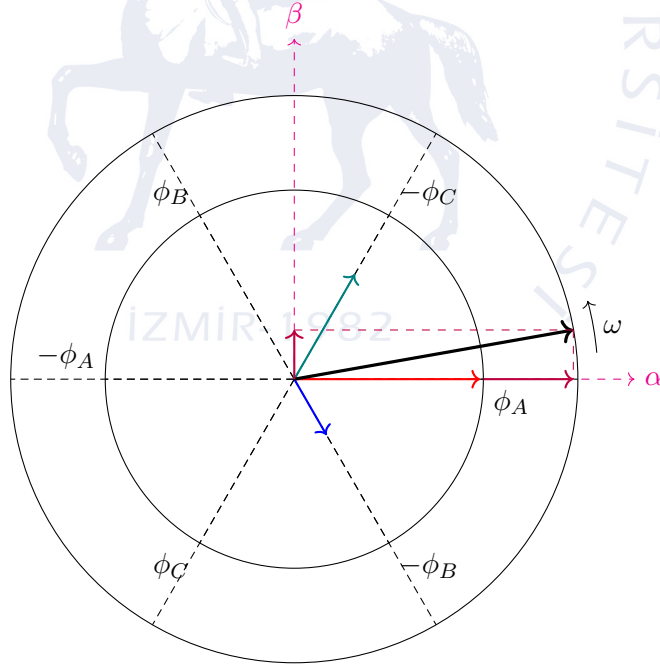


Figure 2.9: Visualization of Clarke Transform

Mathematical and algebraic derivation of Clarke Transformation given below.

$$\vec{V}_r = \vec{v}_a + \vec{v}_b + \vec{v}_c = v_a \hat{a} + v_b \hat{b} + v_c \hat{c} \quad (2.15)$$

$$\vec{V}_r = \vec{v}_\alpha + \vec{v}_\beta = v_\alpha \hat{\alpha} + v_\beta \hat{\beta} \quad (2.16)$$

$$v_\alpha = \vec{V}_r \cdot \hat{\alpha} \quad (2.17)$$

$$v_\alpha = v_a \hat{a} \cdot \hat{\alpha} + v_b \hat{b} \cdot \hat{\alpha} + v_c \hat{c} \cdot \hat{\alpha} \quad (2.18)$$

$$v_\alpha = v_a + v_b \cos(120^\circ) + v_c \cos(240^\circ) \quad (2.19)$$

$$v_\alpha = \left[v_a - \frac{v_b}{2} - \frac{v_c}{2} \right] \quad (2.20)$$

$$\therefore v_a + v_b + v_c = 0 \quad (2.21)$$

$$v_\alpha = \frac{3}{2} v_a \quad (2.22)$$

This coefficient occurs due to winding ratio of 3 phase over 2 phase

$$v_\beta = \vec{V}_r \cdot \hat{\beta} \quad (2.23)$$

$$v_\beta = v_a \hat{a} \cdot \hat{\beta} + v_b \hat{b} \cdot \hat{\beta} + v_c \hat{c} \cdot \hat{\beta} \quad (2.24)$$

$$v_\beta = v_a \cos(90^\circ) + v_b \cos(30^\circ) + v_c \cos(150^\circ) \quad (2.25)$$

$$v_\beta = \left[\frac{v_b \sqrt{3}}{2} - \frac{v_c \sqrt{3}}{2} \right] \quad (2.26)$$

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \begin{bmatrix} 1 & \frac{-1}{2} & \frac{-1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{-\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (2.27)$$

To obtain same magnitudes for mmf, normalization should apply to the transformation after. Coefficient chosen respect to equation 2.22

$$\begin{bmatrix} v_\alpha \\ v_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & \frac{-1}{2} & \frac{-1}{2} \\ 0 & \frac{\sqrt{3}}{2} & \frac{-\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} \frac{2}{3} & \frac{-1}{3} & \frac{-1}{3} \\ 0 & \frac{\sqrt{3}}{3} & \frac{-\sqrt{3}}{3} \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} \quad (2.28)$$

2.3.2 State Vectors and Sectors

In a 3-Phase Inverter topology, there are 8 different states to visualize without shortening the DC source as mentioned in 180° Conduction Section. Applying Clarke Transformation to these states over their phase voltages yields constant state vectors on the 2 Phase Plane.

Also we can derive all six vectors as:

$$\vec{V}_n = \frac{2}{3}V_s e^{j(n-1)\frac{\pi}{3}}, \quad n = 1, 2, 3, 4, 5, 6, 7, 8 \quad (2.29)$$

Table 2.2: State Vectors With Clarke Transform

State	Constant State Vector	$V_{\phi a}$	$V_{\phi b}$	$V_{\phi c}$	V_α	V_β
000	$\vec{V}_0 = 0$	0	0	0	0	0
100	$\vec{V}_1 = \frac{2}{3}V_s e^{j0}$	$\frac{2}{3}V_s$	$-\frac{1}{3}V_s$	$-\frac{1}{3}V_s$	$\frac{2}{3}$	0
110	$\vec{V}_2 = \frac{2}{3}V_s e^{j60}$	$\frac{1}{3}V_s$	$\frac{1}{3}V_s$	$-\frac{2}{3}V_s$	$\frac{1}{3}$	$\frac{\sqrt{3}}{3}$
010	$\vec{V}_3 = \frac{2}{3}V_s e^{j120}$	$-\frac{1}{3}V_s$	$\frac{2}{3}V_s$	$-\frac{1}{3}V_s$	$-\frac{1}{3}$	$\frac{\sqrt{3}}{3}$
011	$\vec{V}_4 = \frac{2}{3}V_s e^{j180}$	$-\frac{2}{3}V_s$	$\frac{1}{3}V_s$	$\frac{1}{3}V_s$	$-\frac{2}{3}$	0
001	$\vec{V}_5 = \frac{2}{3}V_s e^{j240}$	$-\frac{1}{3}V_s$	$-\frac{1}{3}V_s$	$\frac{2}{3}V_s$	$-\frac{1}{3}$	$-\frac{\sqrt{3}}{3}$
101	$\vec{V}_6 = \frac{2}{3}V_s e^{j300}$	$\frac{1}{3}V_s$	$-\frac{2}{3}V_s$	$\frac{1}{3}V_s$	$\frac{1}{3}$	$-\frac{\sqrt{3}}{3}$
111	$\vec{V}_7 = 0$	0	0	0	0	0

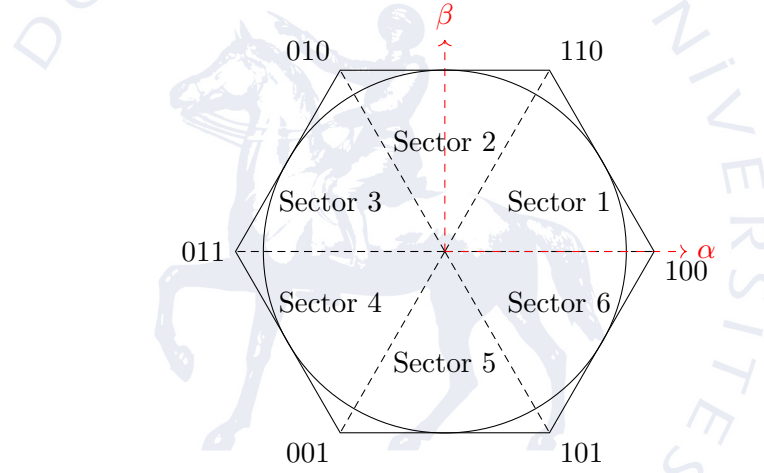


Figure 2.10: Visualization of the State Vectors and Sectors with Clarke Transform

There are 2 zero vectors for states 111 and 000. Having 2 different states which yield to zero vector creates a great opportunity for aligning gate signals. Also since there are constant state vectors on the plane we can define the in-between area as sectors and numerically derive each of them which can be seen in figure 2.10.

2.3.3 Obtaining Gate Signals

There are a couple of ways to obtain gate signals for Space Vector PWM. In this section, Conventional SVPWM and Carrier Based SVPWM are examined.

Conventional SVPWM

Rotating Voltage Vectors can be created by using state vectors according to sector data. For example, the vector is rotating at Sector 1. Vector States: 100, 110 and Zero Vectors are boundaries of Sector 1. So any vector rotating at Sector 1 can be created by using the summation of 100, 110, and Zero Vectors.

So by switching between states we can create the desired vector which showed in figure 2.11. To achieve this in reality, we need a sampling frequency that is much higher than Vector's rotational frequency. During one sampling period, we can derive a mathematical expression for the desired vector.

$$T_s = T_1 + T_2 + T_0 \quad (2.30)$$

For Sector 1:

$$T_s V_r e^{j\theta} = T_1 \frac{2}{3} V_s + T_2 \frac{2}{3} V_s e^{j\frac{\pi}{3}} + T_z \times 0 \quad (2.31)$$

Where:

T_s : Sampling Period

T_1 : Dwell time for first state

T_2 : Dwell time for second state

T_z : Dwell time for zero state

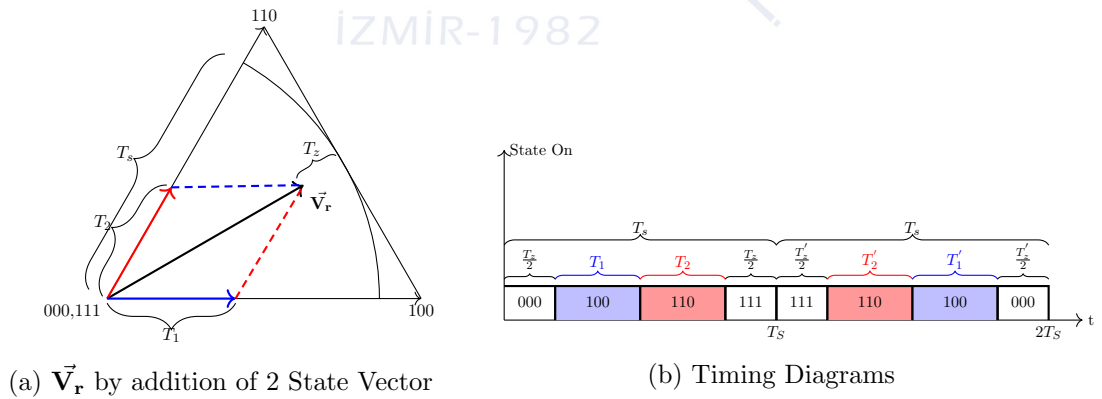


Figure 2.11: Illustrated Vector Addition in SVPWM

To calculate Dwell Times for every sector we can use

$$M = V_r \frac{\sqrt{3}}{V_s} = \sqrt{V_\alpha^2 + V_\beta^2}$$

$$\theta = \sqrt{\arctan \frac{V_\alpha}{V_\beta}}$$

$$T_1 = T_s \times M \times \sin\left(\frac{\pi}{3} \times k - \theta\right) \quad (2.32)$$

$$T_2 = T_s \times M \times \sin\left(-\frac{\pi}{3} \times (k - 1) + \theta\right) \quad (2.33)$$

$$T_z = T_s - T_1 - T_2 \quad (2.34)$$

Where:

M: Modulation Index

k: Current Sector

θ : $\vec{V}_r$'s Angle

V_α , V_β : $\vec{V}_r$'s projectile on α and β axes respectively.

It must be noted that, there are 2 available zero state vectors which are 111 and 000. Our main goal at switching sequence, achieve a minimum number of toggled legs during the transition between state vectors. So having 2 zero vectors makes it possible to change from various vectors to zero vector with one leg toggle. Respect to this philosophy switching sequences for every sector given in table 2.3 below.

Table 2.3: State Switching Sequences

Sector	State Sequence for $2T_s$						
1	000	100	110	111	110	100	000
2	000	010	110	111	110	010	000
3	000	010	011	111	011	010	000
4	000	001	011	111	011	001	000
5	000	001	101	111	101	001	000
6	000	100	101	111	101	100	000

Reference signals can be drawn with respect to the sampling period since we know the conduction duration of every switch for every sector. In table 2.4 conduction duration for legs and in figure 2.12 generated reference signal using this data can be seen. Achieved signals are compared with a carrier, this results with the desired gate signals.

Table 2.4: Conduction Times for High Side Switches Over 1 Period

Sector	Q1 Conduction Time	Q3 Conduction Time	Q5 Conduction Time
1	$\frac{T_s}{2} + T_1 + T_2$	$\frac{T_s}{2} + T_2$	$\frac{T_s}{2}$
2	$\frac{T_s}{2} + T_1$	$\frac{T_s}{2} + T_1 + T_2$	$\frac{T_s}{2}$
3	$\frac{T_s}{2}$	$\frac{T_s}{2} + T_1 + T_2$	$\frac{T_s}{2} + T_2$
4	$\frac{T_s}{2}$	$\frac{T_s}{2} + T_1$	$\frac{T_s}{2} + T_1 + T_2$
5	$\frac{T_s}{2} + T_2$	$\frac{T_s}{2}$	$\frac{T_s}{2} + T_1 + T_2$
6	$\frac{T_s}{2} + T_1 + T_2$	$\frac{T_s}{2}$	$\frac{T_s}{2} + T_1$

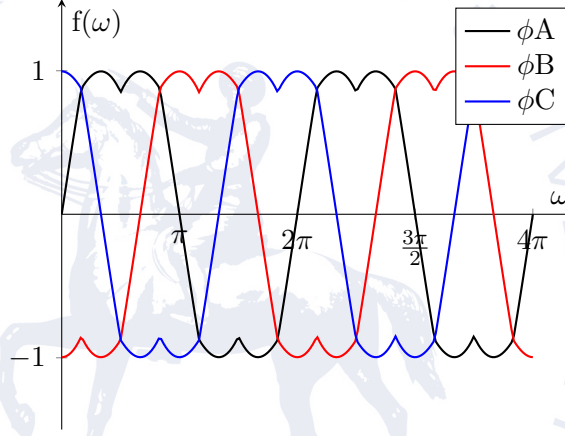


Figure 2.12: Highly Sampled Conduction Time Data Plot

It must be noted that the selection of T_s is critical and plays a huge role in output waveforms. The sampling period must be chosen multiple of 6 since there is 6 defined sector available. This prevents shifting on every 360° scan of the reference vector. Not considering this case in design procedure ends up with uncharacteristic harmonics [6].

Carrier Based SVPWM

Unlike the Conventional SVPWM method to generate gate signals, there is also a method called Carrier Based SVPWM (CBSVPWM). In CBSVPWM no need for Clarke Transformation and dwell-time calculations. Reference Signals are modified and like SPWM compared to the high-frequency carrier signal. Obtained Waveforms are exactly the same as the waveforms obtained from the common method.

In CBSVPWM reference signals are compared to each other continuously. In every sample maximum and minimum point of 3 signal gathered. A new signal can be created with gathered data:

$$f(x) = -\frac{(V_{a|b|c,max} + V_{a|b|c,min})}{2} \quad (2.35)$$

By adding this signal to all sine waves, desired reference signals can be achieved. Example procedure shown in figure 2.13. Also, we can compare these reference signals with signals obtained at Conventional SVPWM. It can be seen they are the same. By applying comparison with a carrier signal, gate signals can be obtained. [7]

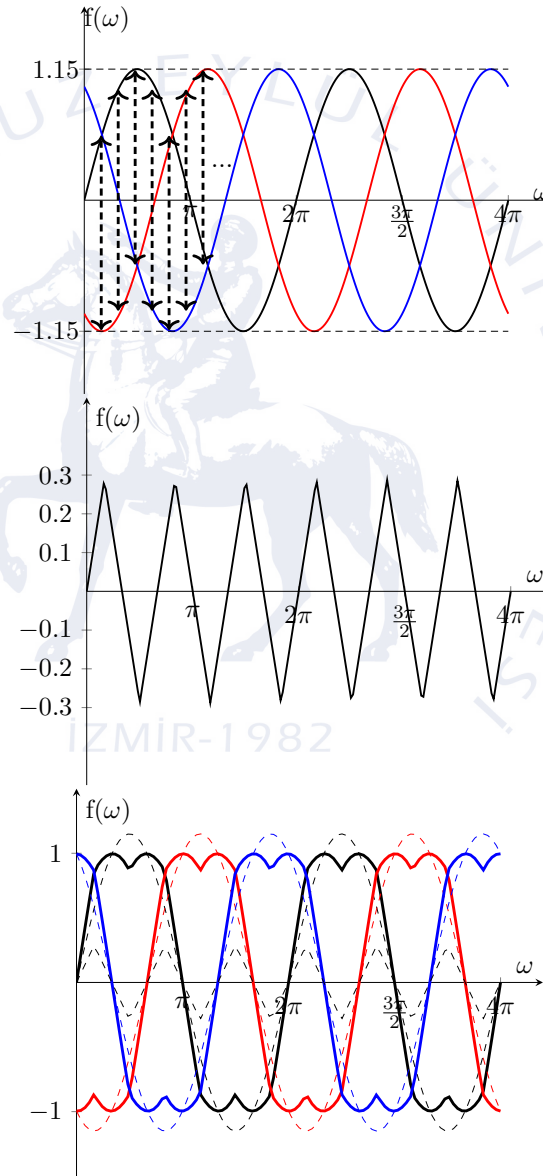


Figure 2.13: Maximum Minimum Detection with Real Time Sampling for 3-Phase System

2.3.4 Modulation Index and Linear Region

If we try to calculate magnitude of a reference vector through β axis. We can calculate magnitude by making geometrical analysis on space vector given figure 2.14 as:

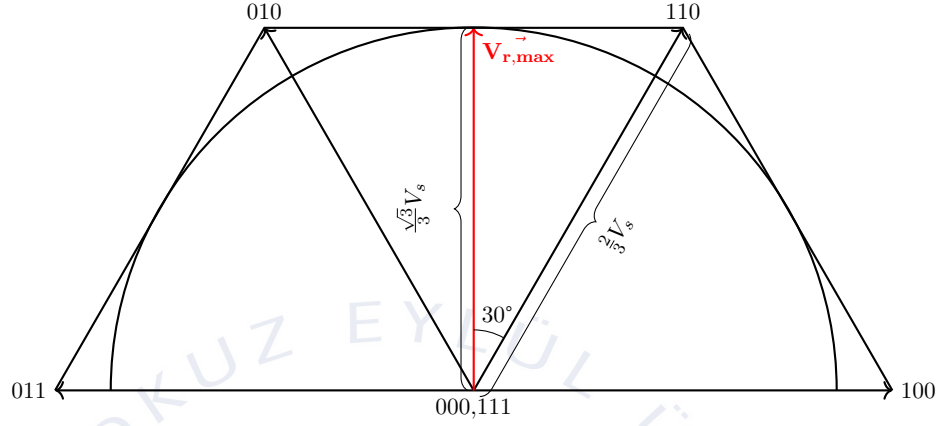


Figure 2.14: Geometrical Derivation of $\mathbf{V}_{r,max}$

$$|V_{r,max}| = \frac{2}{3}V_s \cos(30)$$

$$|V_{r,max}| = \frac{\sqrt{3}}{3}V_s$$

$$\therefore M_{max} = |V_{r,max}| \frac{\sqrt{3}}{V_s} = \frac{\sqrt{3}}{3}V_s \times \frac{\sqrt{3}}{V_s} = 1 \quad (2.36)$$

$$\therefore V_\phi = |V_{r,max}| = \frac{\sqrt{3}}{3}V_s = 0.577V_s \quad (2.37)$$

This implies in the linear region $0 \leq M \leq 1$ and amplitude of the phase voltages is %15 higher than SPWM.

3. Simulation of the System

With the insight of these informations, simulation for Inverter with Space Vector Pulse Width Modulation technique implemented using MATLAB Simulink software in R2021A version. Both for Conventional and Carrier based techniques, same model created and examined. A detailed explanation of the model and blocks can be found under the Simulation Model section. Also generated waveforms and harmonic analysis of them given in Simulation Results section.

3.1 Simulation Model

Open loop model for 50Hz reference vector and 18kHz sampling frequency created by using Simulink which can be seen at the figure 3.1 below. DC supply voltage set at 100V and modulation index adjust to 1 which is the maximum available value for the linear region.

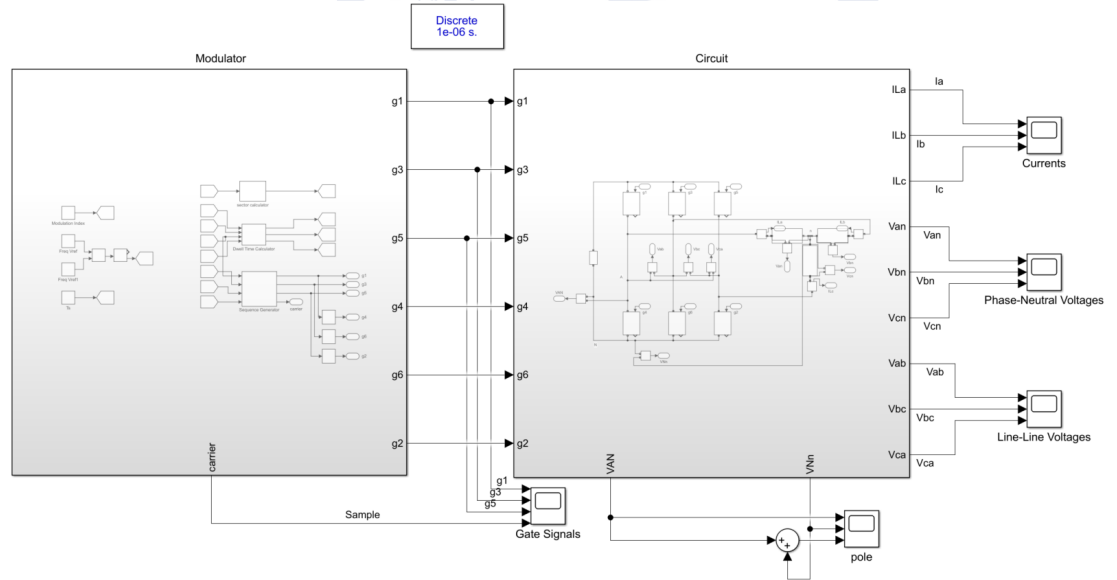


Figure 3.1: Simulink Model

The model is divided into two parts. Hardware section, and Modulator which generates gate signals. For sector calculations, angle data is derived from reference signals. Sampling period, sector data, instantaneous angle, and modulation index were used to calculate dwell times in the submodel. By the conduction table mentioned in table 2.4, gate signals are created using a sequence generator. By using high side signals logically inverted results, low side gate signals are created.

3.1.1 Conventional Modulator

In Modulator block given in figure 3.2 α and β parameters of the reference signal which are sinusoids with 90° phase difference used to derive angle data. Modulation Index(M) and Sampling Period(T_s) data received from initialization file since they are defined as variable parameters to ease simulation.

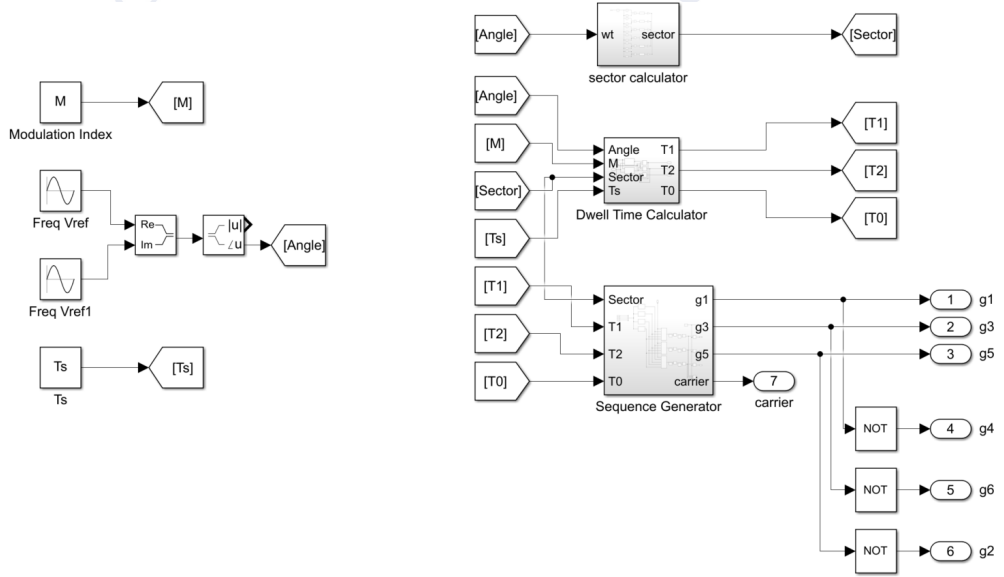


Figure 3.2: Conventional Modulator Structure

Three more submodels called Sector Calculator, Dwell Time Calculator, and Sequence Generator defined. Sector Calculator derives the Reference Vector's current sector in space from its angle data. Dwell Time Calculator, calculates the time that will be spent on each state by using sector data, modulation index, reference signals' angle, and total sampling time. And finally, the Sequence Generator creates switching patterns for the PWM-VSI Circuit's gate signals.

Sector Calculator

The Sector Calculator submodel(given in figure 3.3) simply takes the angle data in radians, converts it to degrees, and derives the sector data using conditional expressions with the sector bound's angle. The output of the logic operator in Simulink must be converted to an integer data type before it can be used in mathematical operations, so the data conversion block is connected to the AND operators' outputs. Summation block used to obtain only one output data. And to prevent critical angles (like 0, and 360 degrees) effect on output data linearization block used. With the help of this block, sector data which are between 1 and 6 are obtained at the outputs since there is 6 sector.

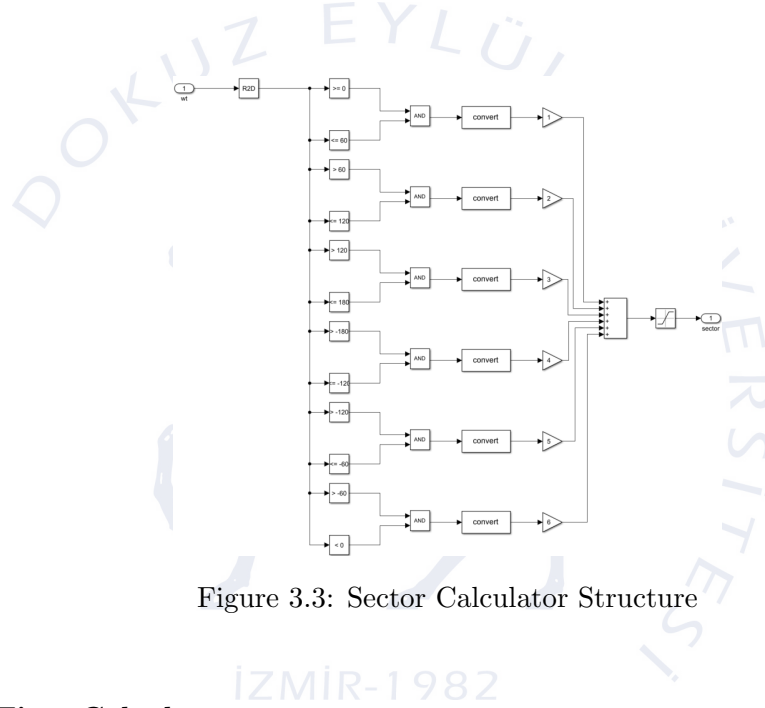


Figure 3.3: Sector Calculator Structure

Dwell Time Calculator

Dwell Times for each vector calculated using equations 2.32, 2.33, 2.34. Upper function block contains formula $\sin(\frac{\pi}{3} \times v - u)$ where u is angle and v is sector data. Multiplying output with sampling period and modulation index gives the dwell times for the first vector. Similarly lower function block contains formula $\sin(-\frac{\pi}{3} \times (v - 1) + u)$. Multiplying the output with the sampling period and modulation index results with the dwell times for the second vector. And dwell time for zero sectors obtained by subtracting these two dwell time parameters from the sampling period.

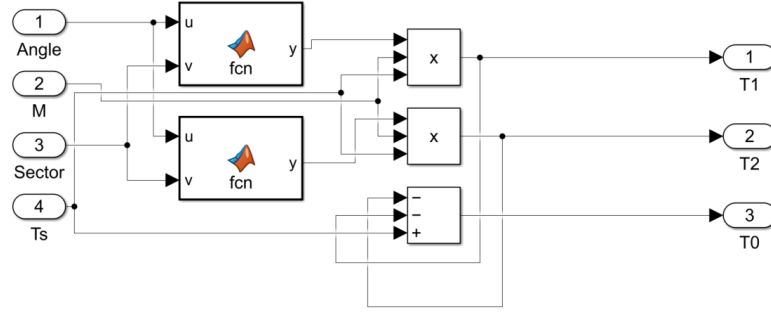


Figure 3.4: Dwell Time Calculator Structure

Sequence Generator

Using the data calculated in the Conventional SVPWM section, switching duration of high side gates for each sector derived. There might be 3 high side gates and 6 sectors but there are only 4 common duration data for all gate signals and for every sector. At the upper left function blocks of the figure 3.5, these 4 common duration data are calculated using dwell time inputs. And also using sector info as a multiplexer's select option, a system has been created to switch between common durations for different sectors. As can be seen in the s1 multiplexer block (for gate 1), sectors 1 and 6 uses $T_1 + T_2 + \frac{T_0}{2}$ duration data, sector 2 uses $T_1 + \frac{T_0}{2}$, sector 5 uses $T_2 + \frac{T_0}{2}$, and finally sectors 3 and 4 uses $\frac{T_0}{2}$.

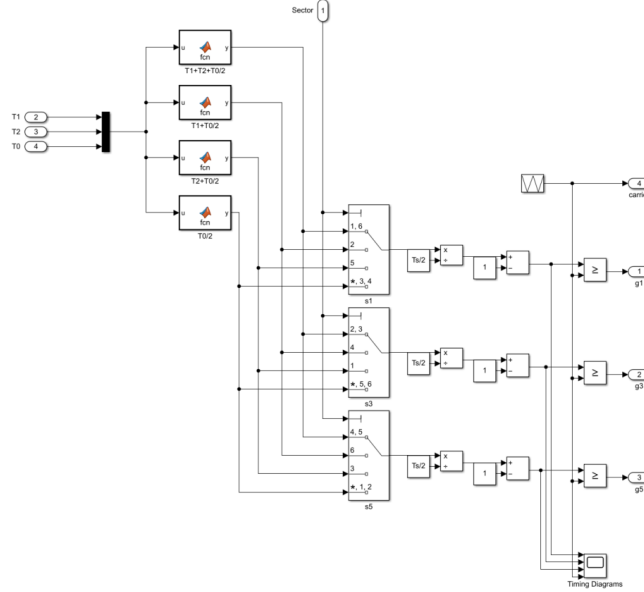


Figure 3.5: Sequence Generator Structure

Since the time data cannot be the negative output of the multiplexer oscillates between 0 and T_s . To normalize that data basic operations are used. Dividing it by T_s results with oscillation between 0 and 1. Multiplication with 2 and subtraction with one finally results in an oscillation between -1 and 1 where amplitude of the data is equal to modulation index. With normalized output (which is data of conduction time for each high side gate) switching patterns can be created easily. Each sampling data are compared with a triangular wave with the same period. Resultant is a switching pattern for gate signals. Also, low side switching patterns of legs derived easily from the logical inverse of high side switching patterns.

3.1.2 Carrier Based Modulator

The model shown in figure 3.6 takes normalized reference signals as inputs and uses the Maximum-Minimum Generator block to create an injection pattern. New patterns goes into the comparison with triangular carrier signal resulting with upper gate signals. Low sides gate signal patterns are derived from the logical inverse of high side switching patterns.

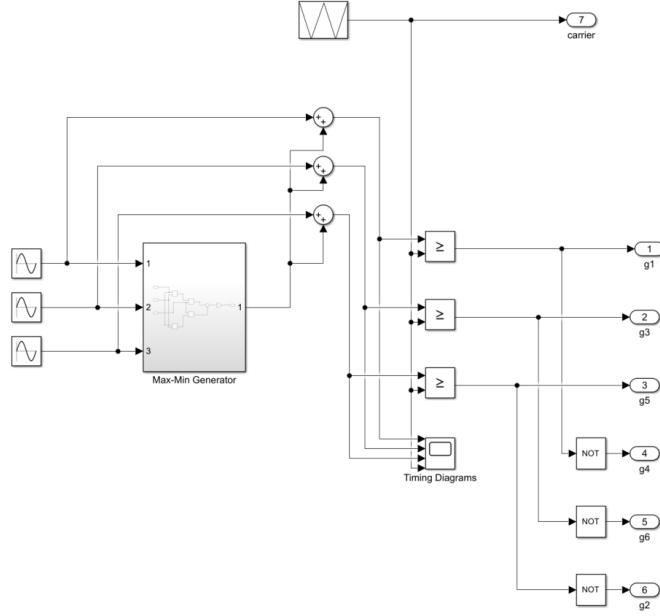


Figure 3.6: Carrier Based Modulator Structure

Max-Min Generator

All 3-Phase reference inputs are put into comparison blocks called "Max" which gives the result as a maximum point which can be seen in figure 3.7. Also, all reference inputs are used to feed a similar system that compares minimum points with the help of the "Min" comparison block. The resultant is used to create an injection signal with the help of equation 2.35.

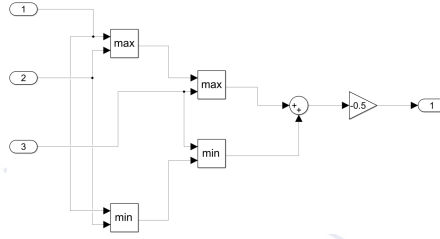


Figure 3.7: Max-Min Generator Block

3.1.3 Inverter Circuit

Inverter Circuit shown in figure 3.8 created by using 100V DC supply, idealized IGBTs and 3-phase RL load. Line to Line, Phase, Line to DC Bus ground and Neutral to DC Bus ground voltages measured with phase current. Measured signals saved to database for Fourier Analysis.

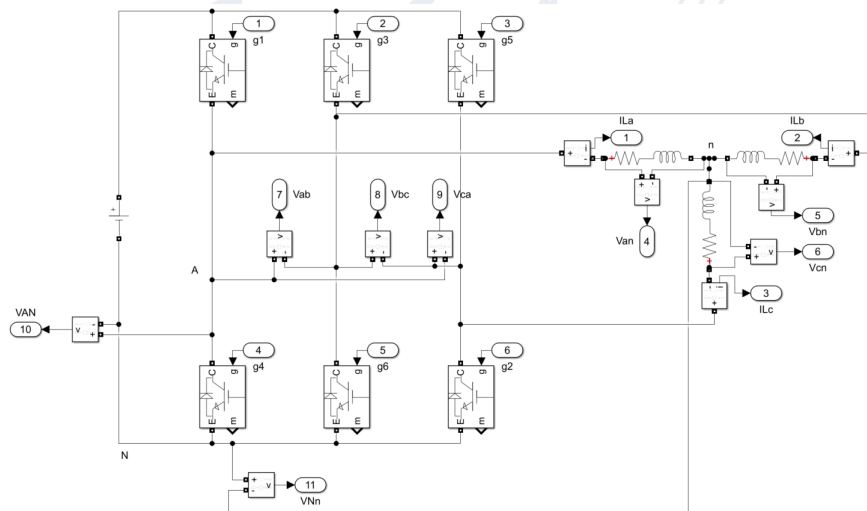


Figure 3.8: 3-Phase PWM-VSI Circuit

3.2 Simulation Results

Results of the simulation model are given below sections. In the first two sections; obtained reference signals, switching patterns, and voltage waveforms are examined. In the FFT Results section harmonic analysis on both pole voltages(V_{AN}), neutral to ground voltages, and phase voltages are shared. Results are compared to the theoretical results found in the Literature Review section.

3.2.1 Gate Reference Signals and Gate Signals

Gate Reference Signals and Gate Signals obtained from Modulator which can be seen in figures 3.9 and 3.10 are identical reviewed ones in the Space Vector Pulse Width Modulation Section figure 2.12. Reference signals have 3rd harmonic components in themselves as expected. Also, it can be seen that, in every sampling, 4 state transition happens to derive a reference vector. From the figure 3.10b it is clearly inference be made that the reference vector in 1st sector since state change happens between zero states and 110 100 states like early examined ones in figure 2.11.

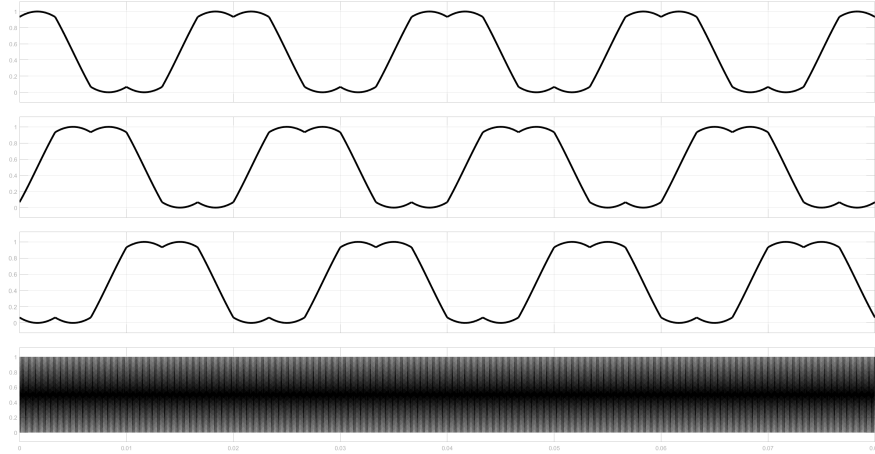


Figure 3.9: Gate Reference Signals

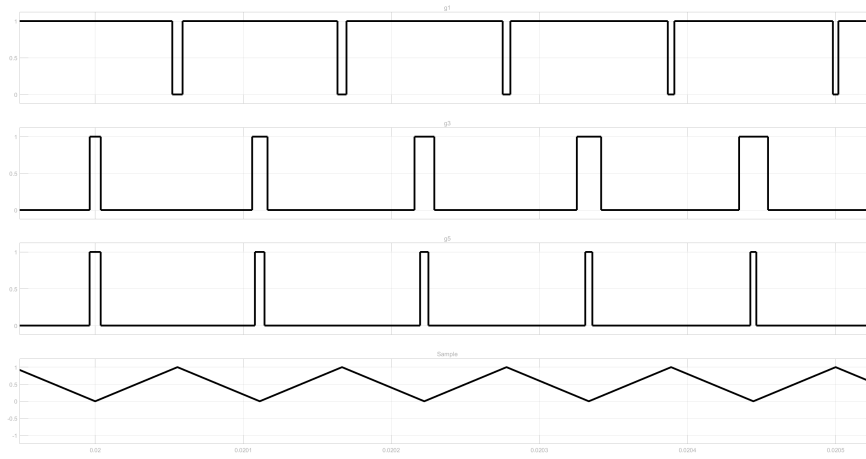


Figure 3.10: Gate Signals in Sector 1

3.2.2 Phase Currents and Phase Voltages

Phase currents are smooth sinusoids with 120° phase difference to each other (figure 3.11). Since RL load acts as a low pass filter and resultant signal reduced from high frequency components which are clearly witnessed in phase currents can be seen in figure 3.12. Also, it must be noted that there is no 3rd harmonic component in the signal even though it has in gate reference signals. Similarly, phase voltages are sinusoids with high frequency switching components.

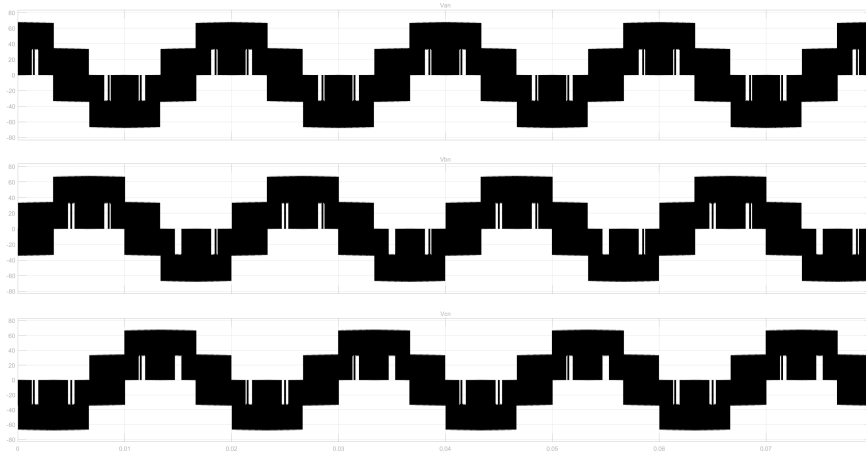


Figure 3.11: Phase Voltages

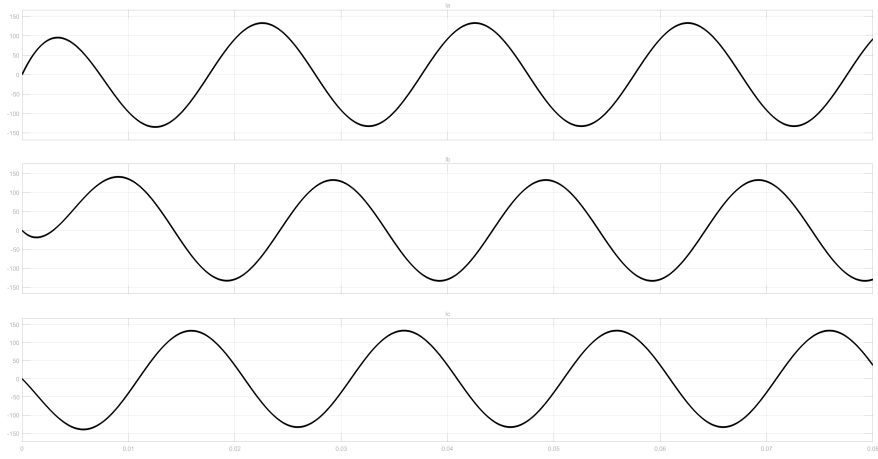


Figure 3.12: Phase Currents

Mathematically phase voltages can be derived using Line to DC Bus ground and DC Bus ground to neutral.

$$V_{\phi} = V_{An} = V_{AN} + V_{Nn} \quad (3.1)$$

Using Pole Voltage V_{AN} and Ground to Neutral Voltage V_{Nn} ; 3rd harmonic components can be analyzed. For this reason sample for these two waveform shown in figure 3.13.

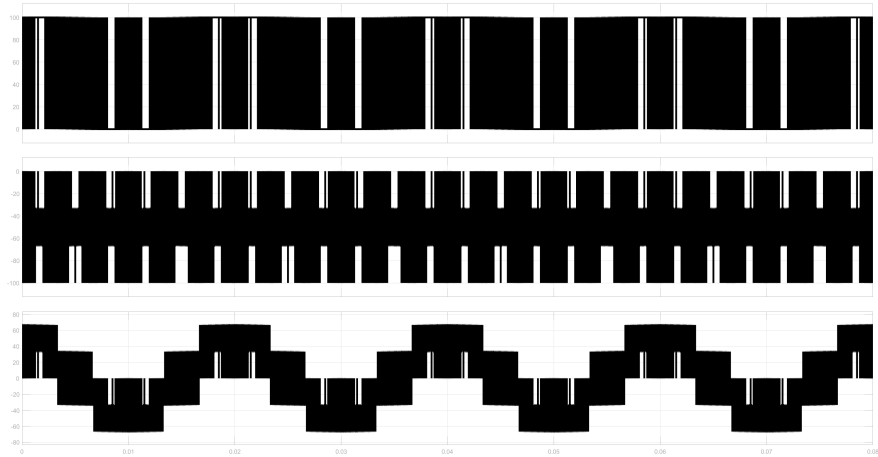


Figure 3.13: V_{AN} , V_{Nn} and V_{An}

3.2.3 FFT Results

FFT Analysis of given signals made by using FFT Analyzer app in Simulink. Phase voltages are examined for clarification on equation 2.37. Also pole voltages and DC Bus ground to neutral voltages are examined for understanding 3rd harmonic components cancellation.

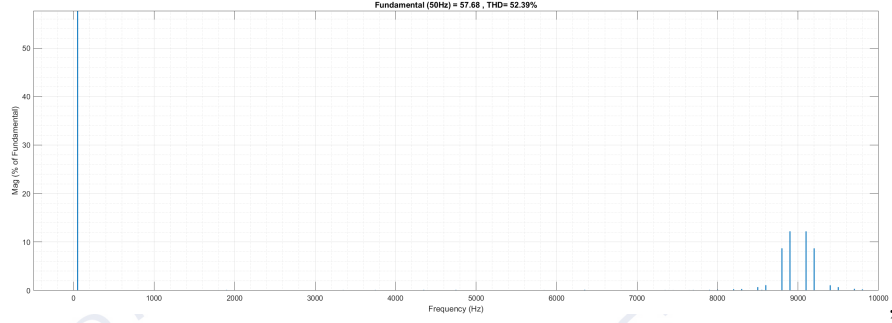


Figure 3.14: FFT Result for Phase Voltage

In FFT Analysis for phase voltages, it can be seen in figure 3.14 fundamental frequency (50Hz in this scenario) reaches %57.7 amplitude for specified base value. This is completely matched with the result found in the Space Vector Pulse Width Modulation section equation 2.37. Also, 3rd harmonic component is zero which can be seen at 150Hz. It must be noted that the signal contains only fundamental frequency and, frequency components near to sampling frequency.

Analysis on V_{Nn} and V_{AN} shows that both signals have 3rd harmonic component with the same magnitude. Which also clear in figures 3.15 and 3.16 there is a phase difference of 180° between signals for their 3rd harmonic. The summation of both signals creates phase voltages as we examined in Simulation Model. So 3rd harmonic component cancels each other due to 180° phase difference. Also, the mathematical expression of this analysis is given below. [8]

$$V_{Nn} = \frac{-1}{3}V_S \sum_{k=1,2,3} d_k \quad (3.2)$$

Where d_k is the states for the upper switches shown in table 2.1.

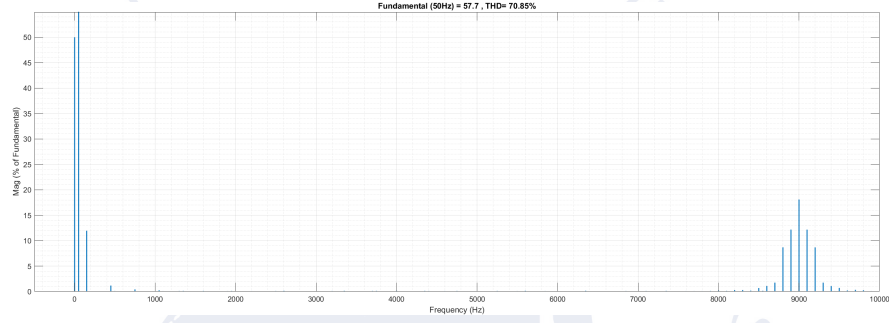


Figure 3.15: FFT Result for V_{AN}

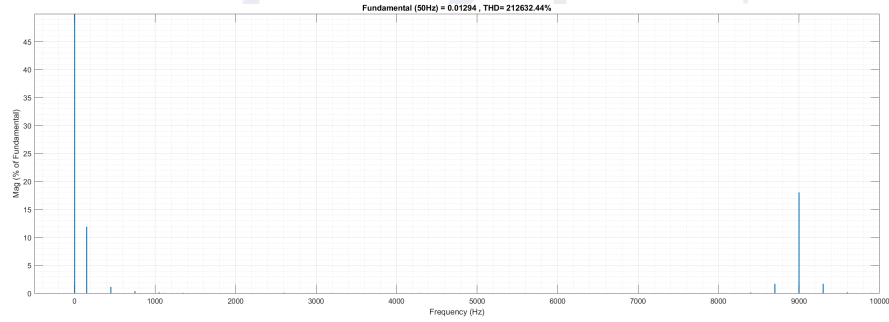


Figure 3.16: FFT Result for V_{Nn}

4. Implementation of the System

In this chapter design process and realization of the system is described in detail. In the implementation, observations based on simulations are considered. Conventional SVPWM method used to generate switching sequence which simulated before.

4.1 Hardware Implementation

To implement the Space Vector PWM modulation technique, the PWM-VSI circuit was designed using KiCAD 5 PCB Design Software. In this section; the design processes of subcircuits, possible drawbacks of the system, and the philosophy behind the design are explained in detail.

4.1.1 Schematic Design of the Circuit

Schematic Design of the Circuit drawn using KiCAD 5 software. Subcircuits are divided into parts with blue dotted boundaries to ease readability. Every division is specially named. In the next sections, all divisions are described individually. The complete drawing can be seen in figure 4.1.

Power Circuitry

Five different Switch Mode Power Supply inputs are taken by using screw terminals. These isolated SMPSs are required for feeding gate drivers. The voltage range of SMPSs is 13V since it is enough to trigger the chosen IGBT gate, according to the datasheet. Also, AC grid is connected to the circuit by screw terminals too. Grid rectified by using KBPC3510 bridge rectifier which is capable of handling inrush currents up to 35 Amperes. Bulk Capacitor for circuit chosen as $750\mu\text{F}$, using equation 4.1 below. Charge current chosen as 1A, Ripple Voltage as 20V, and rectified signal in 100 Hz. The result is $500\mu\text{F}$ but with %50 margin $750\mu\text{F}$ capacitor chosen.

$$C = \frac{I}{f \times \Delta U} \quad (4.1)$$

where;

I: Charge Current, ΔU : Ripple Voltage, f: Rectified Signal's Frequency

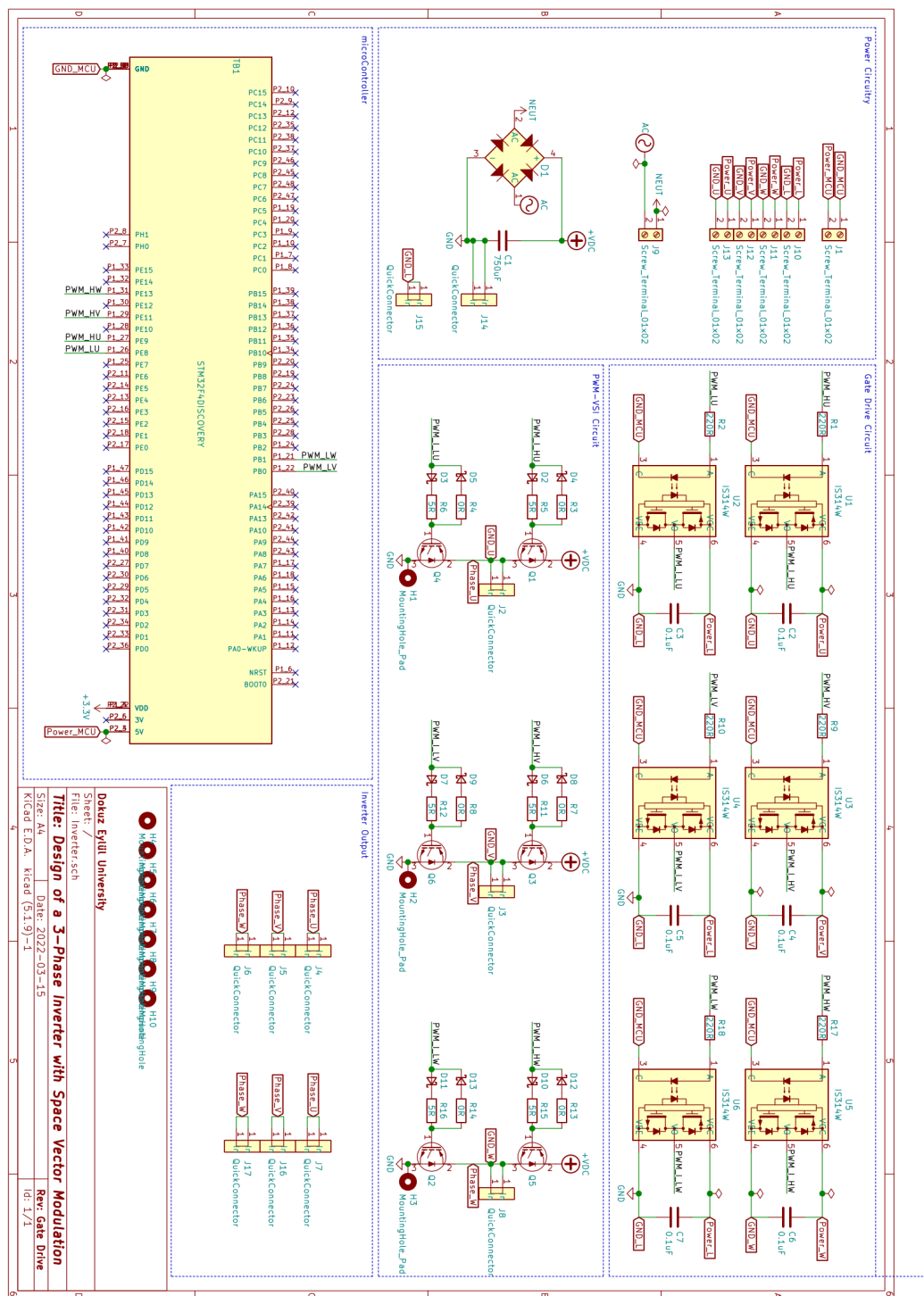


Figure 4.1: Schematic Drawing of the Circuit

Gate Drive Circuit

Due to topology of PWM-VSI circuit gate signals, direct connection of gate signals to the switching semiconductors may end in catastrophic faults. Connecting reference points of the gate signals to the emitters of the IGBT's, results with a short circuit path between low side common ground and high side emitters through Digital Signal Processor ground. Low side switches out of the system in this condition due to short circuit path. In normal gating sequence this causes shorting of DC BUS through high side switches which ends up frying semiconductors, DSP and PCB. In grid without circuit breakers might cause fire hazard too.

To prevent these effects gate driver integrated circuits called IS314W used for optical isolation between DSP and IGBTs' gates. IS314W has enough insulation margins for usage between DC Bus and DSP. IS314W also allows to operate in 10kHz bandwidth which is system's operation frequency. Gate Drive Circuit designed as suggested in its data sheet which can be seen in figure 4.2 below. Since low side switches has common ground in emitter, only one isolated voltage source supplied to the gate drivers connected to them.

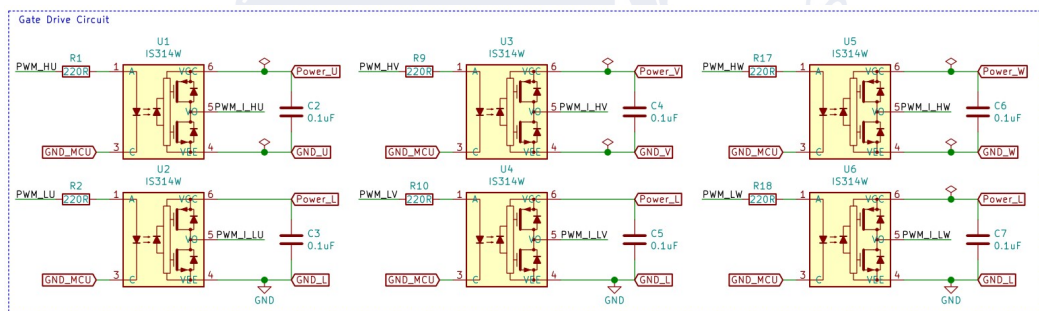


Figure 4.2: Gate Drive Circuit

PWM-VSI Circuit

3-Phase Voltage Source Inverter Circuit is designed using IGBT's as switching semi-conductors. IGBTs called FGH40T65SHDF used in circuit since voltage and current rates are overqualified. Considering possible misfiring due to the Miller Effect, 2 parallel schottky diodes connected to the IGBTs' gates. Charging direction has series 5Ω resistor which is recommended in IGBT's datasheet. Discharging direction has no resistor since discharge duration is much longer than charge duration. Output Voltages has taken between high sides and low sides using screw terminal to ease connection to any cable. Schematic of the circuit can be seen in figure 4.3 below.

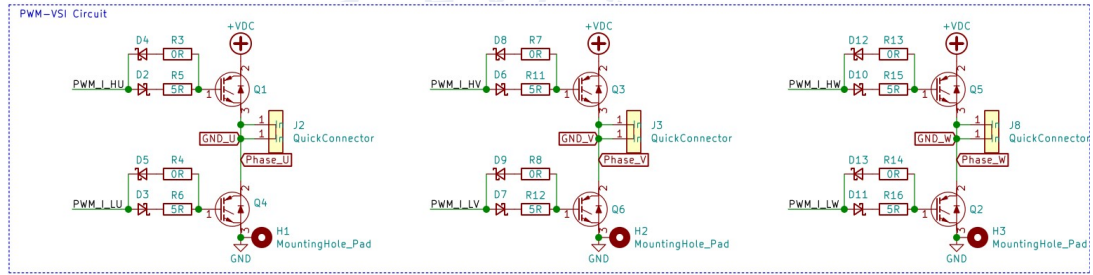


Figure 4.3: PWM-VSI Circuit

MicroController

STM32F407VGT6 microcontroller used for Digital Signal Processing. DSP has a 168MHz clock frequency which is high enough for the application. Also, advanced timer features like 3-Channel complementary PWM output and dead-time insertion influenced the DSP choice.

20kΩ potentiometer connected to the microcontroller for analog readings to get desired frequency inputs of the system. Also, 6 Output signals generated and connected to the gate drive circuit. The generation of the signals explained in detail in the Software Implementation of the System section.

4.1.2 Printed Circuit Board Design of the Circuit

Printed Circuit Board Design of the Circuit drawn using PCB Layout Editor sub-software in KiCAD 5. The resultant design can be seen in figure 4.4. In the design process, IPC-2221 standards are considered for clearance between copper traces. Also considering the amount of currents to be drawn between nodes, copper trace widths chosen with 20°C thermal margin.

In this topology, diode bridge and IGBTs are under thermal stress. In continuous operation, these semiconductors may heat up to critical levels. To prevent this heatsinks are used for these semiconductors. In the design process common heatsink for the high side and also another common heatsink for the low side are used. The flange of the IGBT has an electrical connection with the collector which is mentioned in the datasheet. To prevent any connection between collectors of the IGBTs, thermal isolation pads are used. PCB created with the ironing method. 3d preview of the PCB can be seen in figure 4.4 below.

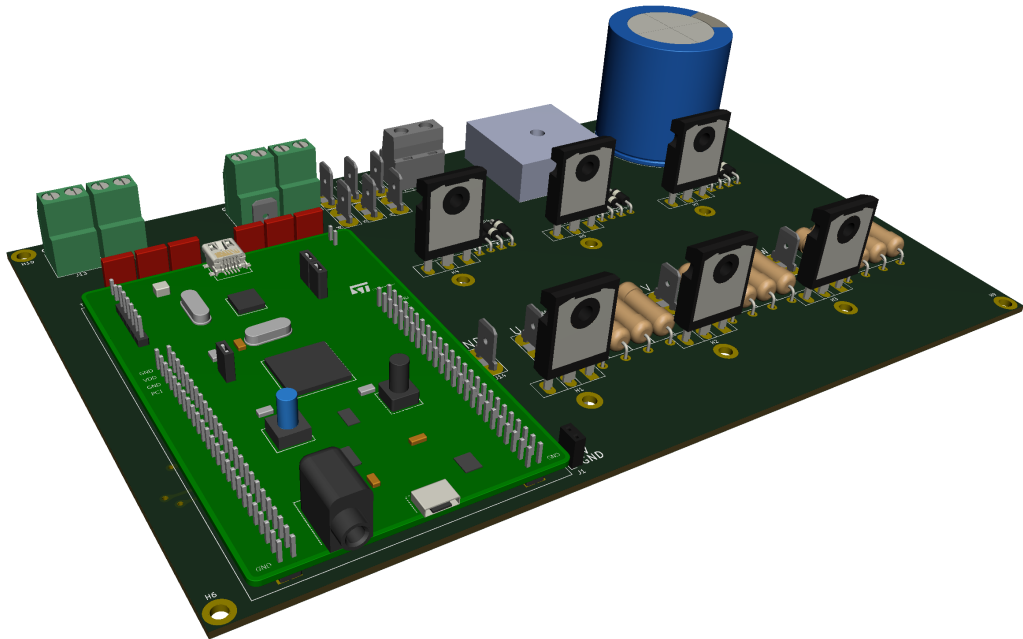


Figure 4.4: 3d Preview of the PCB

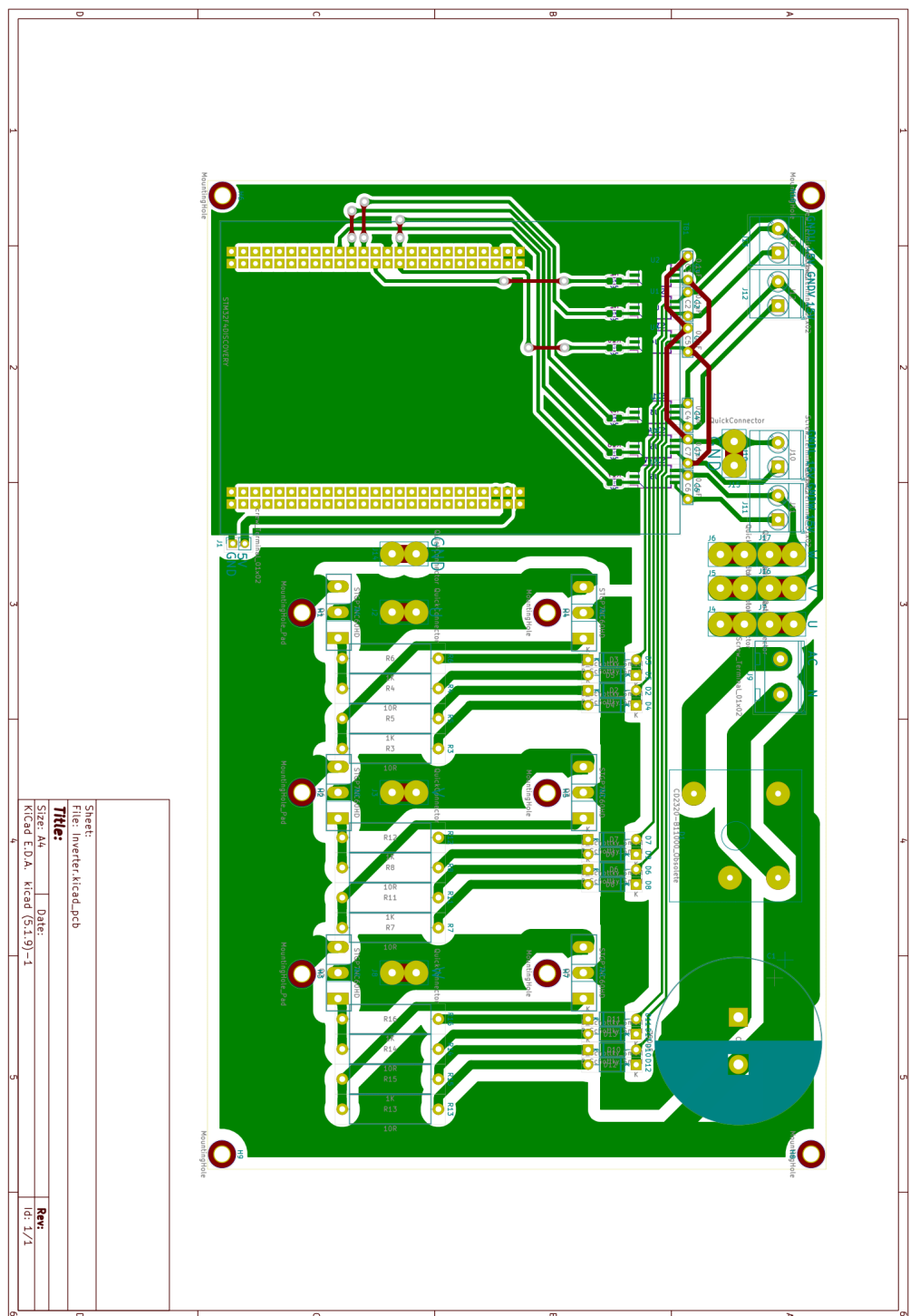


Figure 4.5: PCB Drawing of the Circuit

4.2 Software Implementation

STM32F407VGT6 microcontroller based software designed to generate Space Vector PWM gate signals. In this section; the design processes of software and the philosophy behind the design are explained in detail.

Using an advanced timer peripheral of the DSP made 3-Phase PWM generation uncomplicated. Complementary output feature used to generate low side gate signals. Also, dead time module is used to prevent any overlap between high side and low side switches. Dead time duration was selected by observing switching durations of the IGBTs. Test results shared in figure 5.1 in Experimental Results and Discussions section.

A timer interrupt routine designed for $100\mu\text{S}$ gaps, which implements a 10kHz sampling ratio of the system. Interrupt routine performs necessary calculations for the open loop system at the desired frequency which is taken from the user by ADC peripheral. Using shadow registers, compare registers of the PWM output updates itself synchronously. In every sample, calculations of the next sample's dwell times are performed using the Conventional SVPWM method.

Every cycle uses normalization arithmetic for the modulation index to adjust the flux of the load concerning the output frequency. Frequencies greater than the boundaries of the modulation index results in flux weakening since the algorithm limits the modulation index at a maximum of 1 point to prevent overmodulation.

An external hardware interrupt is designed for activating or deactivating PWM outputs for safety concerns. Otherwise switching waveforms may be seen on the output terminals and this can result in unintentionally driving motor or other loads. The flowchart of the system can be seen in figure 4.6 below.

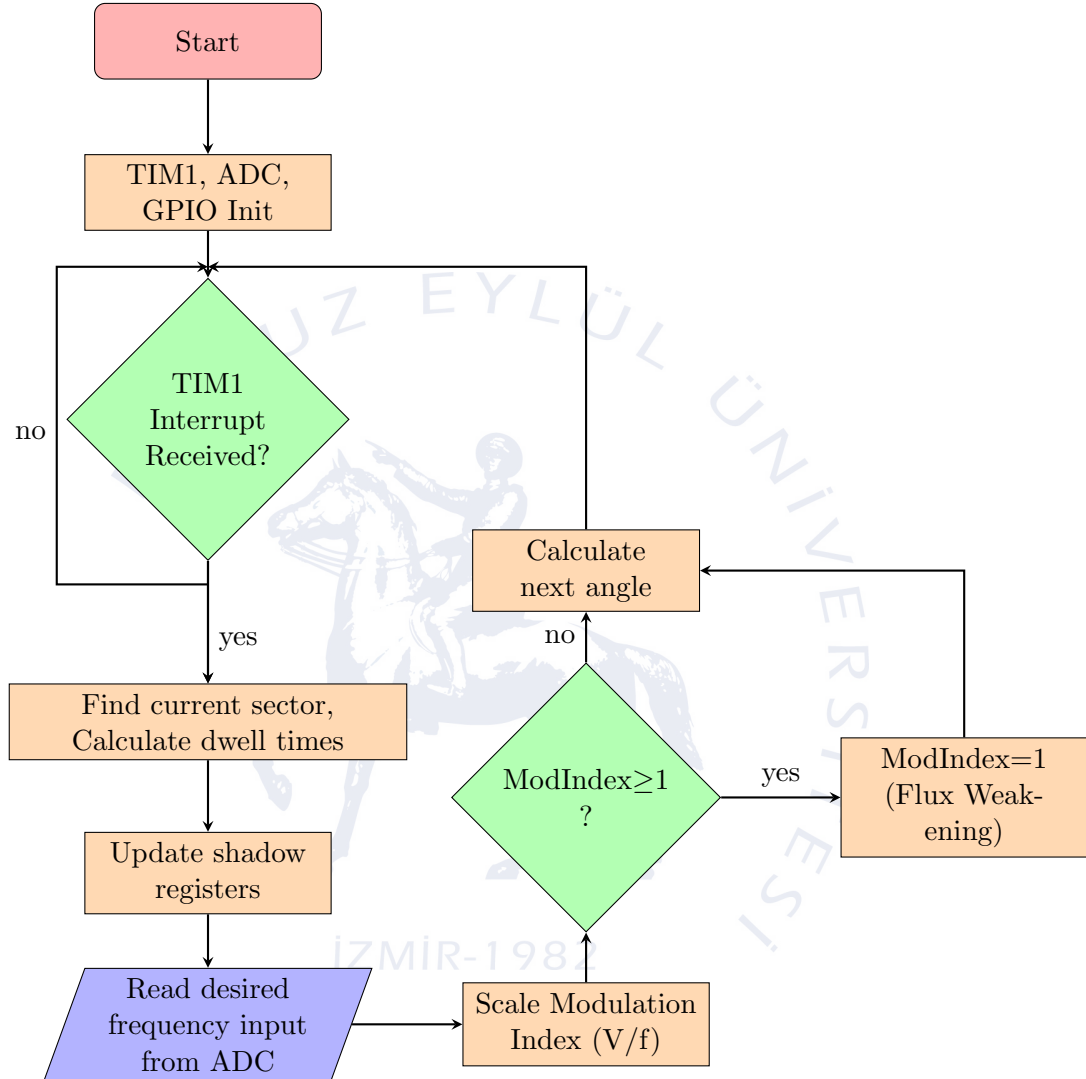


Figure 4.6: Flowchart of the Software

5. Experimental Results and Discussions

In this chapter detailed explanation of the experimental process is shared. During experiments, plenty of hardware tests done to adjust software parameters. Hardware components' operability confirmed. Experimental setup with the asynchronous motor as load used after confirmations. All data measured by oscilloscope and exported to .csv data format to perform Fourier Analysis using MATLAB.

5.1 Hardware Results

After the confirmation of every element's operability, rise and fall time of the IGBTs' gate signals are measured as can be seen in figure 5.1. Using that data, deadtime durations are implemented in software to prevent any overlap in legs.

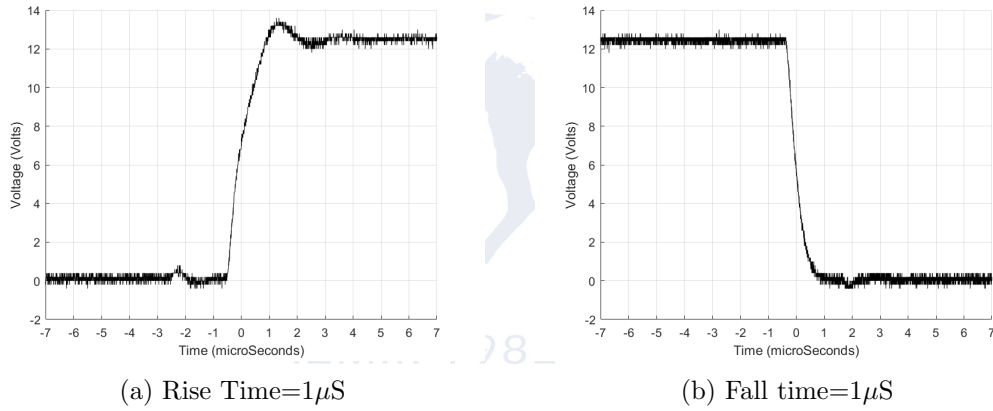


Figure 5.1: Rise and Fall time for IGBTs' Gate

5.2 Software Results

Using the rise and fall time data of the IGBTs' gate signals, implementation of the correct deadtime durations has done. The resulting switching signals created by Digital Signal Processor under the low-pass filter can be seen in figure 5.2 below.



Figure 5.2: Switching signals for one gate after applying low-pass filter

It is clear that obtained waveform matches with the theoretic and simulated waveforms which mentioned earlier in figures 2.12 and 3.9. Also, the waveform includes a distinguishable third harmonic pit at maximum and minimum points which is well expected.

5.3 Experiment Setup

Confirmation of the correct operation of both hardware and DSP directs to the experiment under load. In experiment setup; AC signal for DC Bus input, obtained using a variac. 4 SMPS used to feed gate drivers to ensure isolation.

Load choosen as asynchronous motor with no load condition with empty shaft. Motor used in star connection. According to the nameplate of the motor, star connection has 1.28A nominal current with 380V line-to-line input at 50Hz frequency. Nameplate of the selected AC motor can be seen at figure 5.3 below.

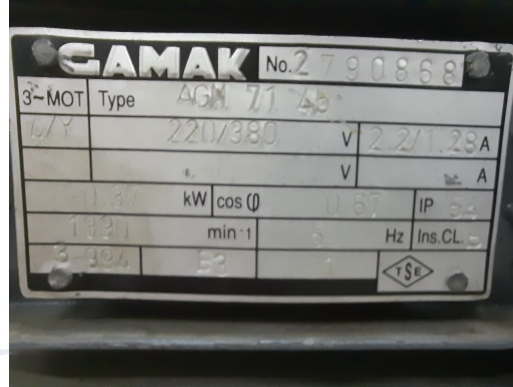


Figure 5.3: Nameplate of the AC motor used in experiments

Experimental setup demonstrated in laboratory photographed which can be seen at figure 5.4 below. Variac at the upper left corner of the desk, motor on its left, and SMPSs next to them can be seen clearly. Oscilloscope with differential probe used to measure output waveforms.

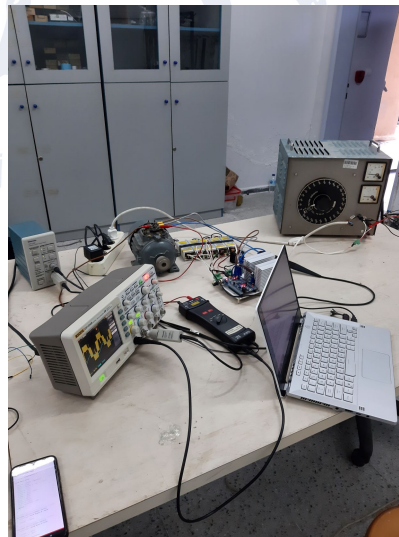


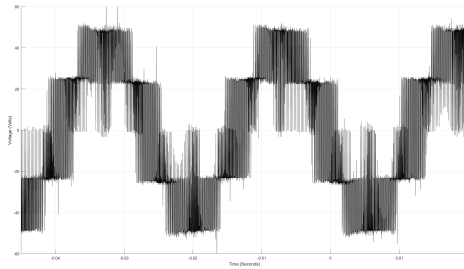
Figure 5.4: Experimental Setup

5.4 Experiment Results and Comparison with Theoric Calculations

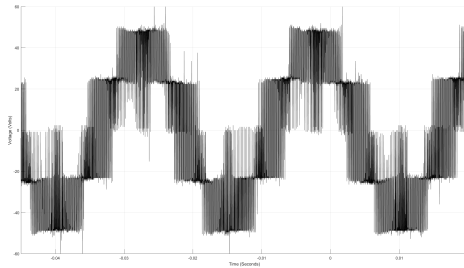
In this section output waveforms for both phase voltages, phase currents, and line-to-line voltages are shared respectively. In the first part, all parameters are shared for every phase and lines for a specific frequency. The next part includes data for different frequency values to observe the system in higher or lower ranges.

5.4.1 Results at 39Hz

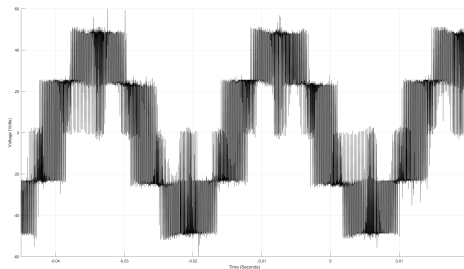
In this setup system is fed by 55V RMS AC which ends up with 75V DC Bus voltage. Motor fed by 39Hz signal which uses 75V DC Bus. Phase voltages, phase currents and line-to-line voltages shared respectively which can be seen on figures 5.5 5.6 5.7below.



(a) Phase U



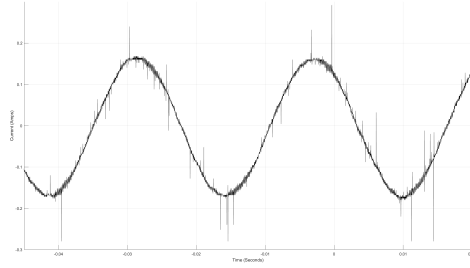
(b) Phase V



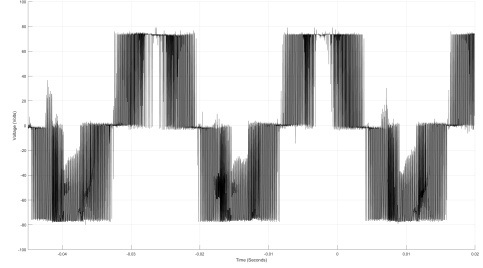
(c) Phase W

Figure 5.5: Phase Voltages at 39Hz

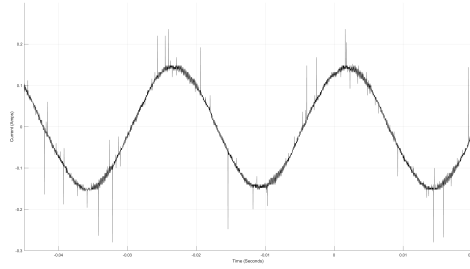
Maximum phase voltage equals to the 50 volt which is $\frac{2}{3}$ times 75V DC Bus as expected and normalized phase value is equal to 43V which is 0.57 times DC Bus which also expected and mentioned in 2.37 earlier. Also, waveforms match the simulation results visually.



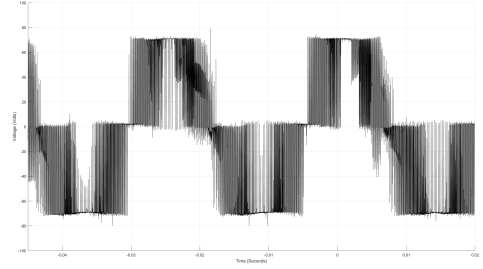
(a) Phase U



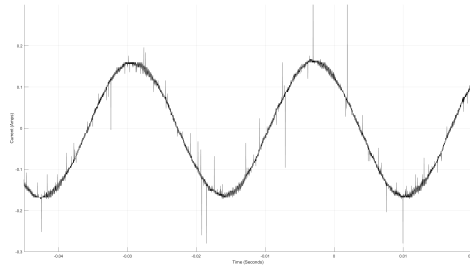
(a) Line AB



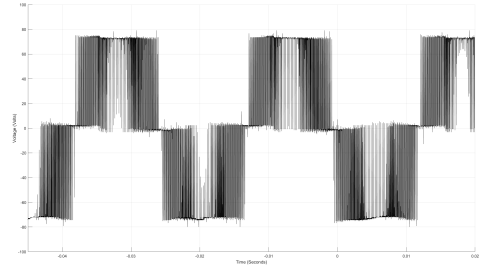
(b) Phase V



(b) Line BC



(c) Phase W



(c) Line CA

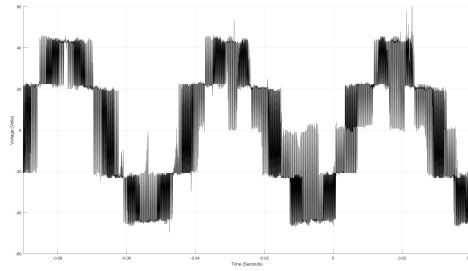
Figure 5.6: Phase Currents at 39Hz

Figure 5.7: Line-to-Line Voltages at 39Hz

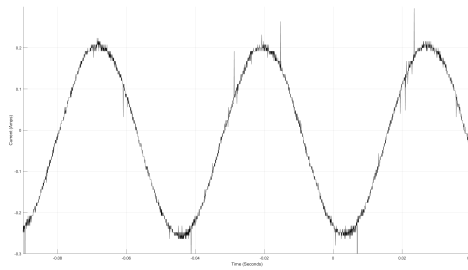
Phase currents are filtered since the motor acts like a low-pass filter for them which is also observed in the simulation part and can be seen in figure 3.12. Equality on amplitudes of the currents indicates the balance of the system. Line-to-Line voltages can be seen on figure 5.7 above.

5.4.2 Operation at Various Frequencies

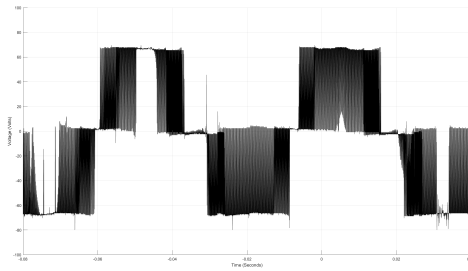
Using the same setup used in the earlier experiment, the operation was observed for various frequencies. The resultant response was seen in motor speed. In lower frequencies, the motor operated at low speed, the exact opposite in the higher frequencies. Measurements taken at 21Hz and 59Hz shared in figures 5.8 and 5.9 below.



(a) Phase Voltage

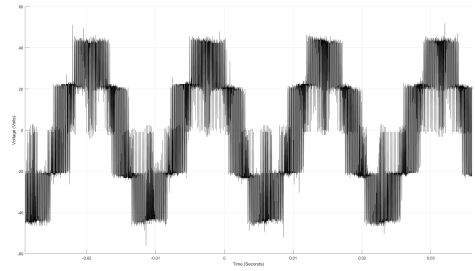


(b) Phase Current

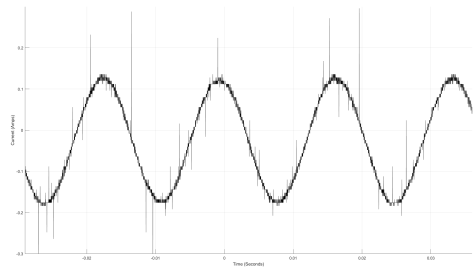


(c) Line-to-Line Voltage

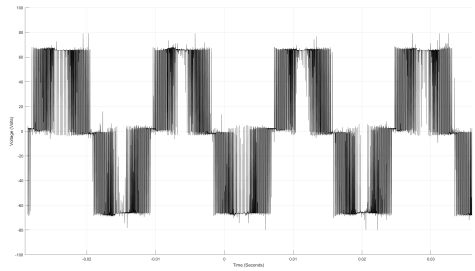
Figure 5.8: Operation at 21Hz



(a) Phase Voltage



(b) Phase Current



(c) Line-to-Line Voltage

Figure 5.9: Operation at 59Hz

5.5 FFT Results of the Measurements

FFT Analysis of phase outputs made by using the FFT Analyzer app in Simulink. Phase voltages are examined for clarification on equation 2.37. In FFT Analysis for phase voltages, which can be seen in figure 5.10 below, 3rd harmonic cancellation occurs as expected and discussed in the FFT results of the simulation. Only high frequency components at levels near 10kHz are visible due to the selected switching frequency.

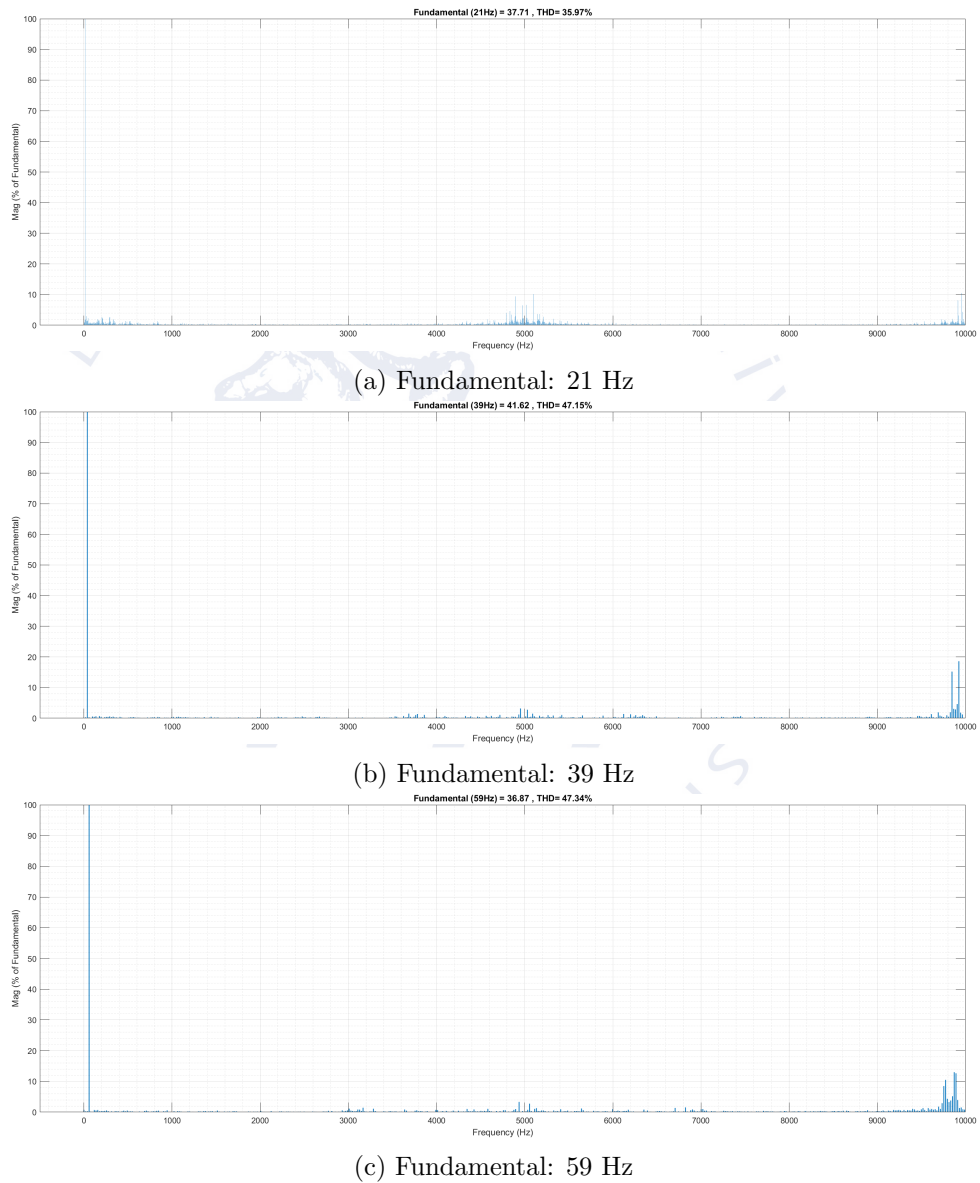


Figure 5.10: FFT Results in shape of Bar Graph

6. Cost Analysis

Economical cost of the demonstration and the system's effect on social life is given below.

6.1 Economical Costs

Cost analysis has been made by considering the system described in the Implementation of the System chapter. To supply DC voltage to VSI Circuit high current rated rectifier and high capacitance bulk capacitor are used. As DSP, STM32F407VG μ controller chosen. Discrete IGBT's FGH40T65SHDF used in VSI circuit. To apply gate signals to IGBT Optocoupler Isolation is needed and IS314W single gate driver is used for this purpose. Also, 4 isolated Voltage sources absorbed by using 13V SMPSs. The cost table including all parts can be seen at table 6.1.

Table 6.1: Cost Table

Part's Name	Type	Number	Notes	From	Price
StM32F407VGT6	MCU	1		OZDISAN	263TL
FGH40T65SHDF	Discrete IGBT	6	600V-15A	OZDISAN	460TL
IS314W	OptoCoupler	6	1MHz	OZDISAN	72TL
KBPC3510	Bridge Diode	1	35A	OZDISAN	26TL
Miscellaneous	Resistors Caps			OZDISAN	50TL
AL20A681DF400	Bulk Capacitor	1	680uF-400V	OZDISAN	40TL
ZHS-1181ASD100MM	Heatsinks	2		OZDISAN	120TL
SMPS	Power Supply	4	15V	OZDISAN	320TL
Total Price					1351TL

6.2 Environmental, Political and Social Costs

The project and the system have no environmental damage. And since the utilization of the DC Bus is much higher than other techniques, using this system to drive an AC motor is much more beneficial. Also, in the social aspect system is beneficial and technique is a milestone in the motor drive area.

7. Conclusion

In conclusion Space Vector PWM requires much more tedious calculations for inverter drive compared to other techniques, but the result is superior to the others. DC Bus utilization is maximum mathematically available for the linear region and also has lower harmonic distortion in lower frequencies.

In this project theoretical analysis was made and shared on SVPWM to be more understandable. Simulation for system performed to obtain reference results for easier implementation of the system. Simulation results and measurements match with theoretical values. Reference Signals and Gate Signals match the theoretical ones. There are no uncharacteristic harmonics due to the correct selection of sampling period also DC bus utilization in %57 levels. Using both Conventional and Carrier Based methods results in the same waveforms which also clarified in Simulation Model and FFT results.

In real-life implementation of the system, Conventional Method used since it was easier to design using Digital Signal Processor. The design process of the system is shared in detail. The resultant waveforms completely match the simulation results. Also, physical observations on AC motor, which is used as load, clarifies smooth operation.

To improve and develop the project further, a closed loop system might be added to the system. Observation of the rotor speed in continuous time makes torque control possible. Also, that kind of system autocorrects itself without needing any operator for various loads on the shaft.

8. References

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